

The Hodge Conjecture by Exclusion of the Empty Cycle-Fiber Endpoint

An AASC Endpoint-Rigidity Proof for Rational Hodge Classes via Cycle-Fiber Occupation

Amos Jay Maley

June 2026

Audit-grade endpoint-exclusion edition

Abstract

This manuscript gives an AASC endpoint-exclusion proof of the Hodge Conjecture on the fixed Hodge/cycle carrier. The proof does not construct algebraic cycles by a new algebraic-geometric method and does not redefine Hodge classes, Chow groups, rational coefficients, the cycle-class map, or algebraic cycles. The official endpoint is the classical assertion that every rational Hodge class on a smooth projective complex variety has nonempty rational cycle-class fiber. The proof has one nonclassical dependency: official endpoint-defeating counterexample force is theorem-bearing endpoint use. Once that dependency is granted, the remaining argument is target instantiation plus route exhaustion on the fixed Hodge/cycle carrier.

For a fixed smooth projective complex variety X , codimension p , and rational Hodge class $\alpha \in \text{Hdg}^p(X)$, the positive endpoint image is the occupation of the cycle-class fiber

$$F_X(\alpha) = \{Z \in \text{CH}^p(X)_{\mathbb{Q}} : \text{cl}_{\mathbb{Q}}(Z) = \alpha\}.$$

The positive branch is $F_X(\alpha) \neq \emptyset$. The bare negative branch is $F_X(\alpha) = \emptyset$. Empty-fiber content is lawful as raw mathematical content, as obstruction or nonmembership content, and as ordinary negative theoremhood. It becomes AASC-relevant only when used to defeat the official Hodge endpoint on the same fixed Hodge/cycle carrier. Under such use, empty-fiber counterforce is endpoint use; endpoint use instantiates the kernel of reference, standing, admissibility, and irreversibility; the kernel forces the A^+ consequence layer; and A^+ excludes independent same-domain endpoint-status discriminators.

The native empty-fiber counterforce route is exhausted into bridge-neutralized status, support-only status, bookkeeping, carrier/domain shift, or the independent cycle-fiber discriminator $D_{\text{fib}}(X, p, \alpha)$. In the live exact-complement reductio subproof, the first four routes are unavailable: bridge-neutralization contradicts the live empty-fiber assumption, support-only and bookkeeping lack endpoint counterforce, and domain shift leaves the fixed Hodge endpoint. Thus the local negative countercase induces $D_{\text{fib}}(X, p, \alpha)$, while endpoint use supplies $N_{\text{fib}}(X, p, \alpha)$ and hence $\neg D_{\text{fib}}(X, p, \alpha)$. The contradiction discharges the local empty-fiber assumption. Since X, p, α were arbitrary, every rational Hodge class has nonempty cycle-class fiber. This is the Hodge Conjecture.

Contents

1	Mathematical-status lock, proof-class lock, and front-loaded dependency lock	3
2	Read this first: endpoint proof spine	3
3	Official Hodge endpoint and target instantiation	4
4	Cycle-class carrier and model adequacy	5

5	Kernel, A+, ATS, and UEAP replay for Hodge endpoint use	6
6	Counterexample force as endpoint use	7
7	Structural-to-cycle-fiber equivalence bridge	9
8	Single cycle-fiber endpoint interface	10
9	Native empty-fiber counterforce exhaustion	10
10	Cycle-fiber separator and discriminator closure	17
11	Local reductio and universal Hodge endpoint	18
12	Ordinary practice, route tests, and incompleteness firewall	19
13	Exhaustion, weakening resistance, and denial burden	20
14	Conclusion	21
A	Theorem ladder and dependency audit	22
B	Anti-circularity and hostile-misreading audit	22
C	Route-locus exhaustion matrix	22
D	Notation ledger	23
E	Corpus support ledger	23
F	Lean4 Hodge Endpoint Audit Appendix	25
G	Hostile-referee audit expansion and replay matrix	31
H	Faithful counterexample worksheet	46
I	Expanded stress-test capsules and report-preservation matrix	46
	References	80

1 Mathematical-status lock, proof-class lock, and front-loaded dependency lock

Mode declaration. This manuscript is written in AASC endpoint-exclusion mode. AASC is used as a mathematical constraint formalism: it classifies fixed-domain endpoint use, report standing, admissibility, lawful redescription, and forbidden same-domain discriminator import. It is not an analytic cycle-construction method.

Proof-class lock. The proof class is kernel-forced empty cycle-fiber endpoint-counterforce exclusion. It is not a Lefschetz theorem, not a motivic construction, not an Abel-Jacobi or normal-function argument, not a proof from the standard conjectures, not a deformation or spread argument, and not a computation. Classical Hodge-theoretic objects enter to identify the official carrier and endpoint. The branch-exclusion mechanism is AASC-internal.

Front-loaded dependency lock. This manuscript has one nonclassical dependency: official endpoint-defeating counterexample force is theorem-bearing endpoint use. A bare object, formula, theorem, or local assumption with empty-fiber content is not endpoint use merely by being written. But when that content is used to defeat the official Hodge endpoint on the same fixed carrier, the mathematical act fixes target, carrier, branch, reportability, downstream reuse, branch exclusion, act-time finality, and lawful-redescription-stable endpoint role. That act is endpoint use. Once this dependency is granted, the remaining proof is target instantiation plus route exhaustion.

Immediate denial fork. A same-mode objection must therefore do one of the following: deny that official counterexample force is endpoint use; deny that endpoint use instantiates the kernel and the A^+ consequence layer; deny that algebraic readout is exactly cycle-fiber occupation; deny native empty-fiber counterforce exhaustion; deny the no-independent-discriminator closure; or produce a fifth endpoint role for same-carrier empty-fiber counterforce.

Lean support boundary. The Lean4 archive associated with this manuscript is audit support for the AASC/Hodge endpoint-role surface, not a replacement for the manuscript proof and not a first-principles formalization of algebraic geometry. Its scope is the fixed-carrier endpoint-use route, the empty cycle-fiber separator, the induced fiber discriminator, the no-independent-discriminator closure, and the represented endpoint closeout. Classical Hodge theory continues to supply the native carrier vocabulary: smooth projective complex varieties, codimension, rational Hodge classes, rational Chow groups, and the rational cycle-class map. The release repository and Zenodo record are cited in Appendix F.

2 Read this first: endpoint proof spine

For fixed X, p, α , write

$$\text{StandHdg}(X, p, \alpha) \iff \alpha \in \text{Hdg}^p(X), \quad \text{ReadAlg}(X, p, \alpha) \iff F_X(\alpha) \neq \emptyset.$$

The local proof opens the exact-complement counterforce $[F_X(\alpha) = \emptyset]_i$ against the target $F_X(\alpha) \neq \emptyset$. The proof route is

$$\begin{aligned} [F_X(\alpha) = \emptyset]_i &\Rightarrow \text{LocalEmptyFiberCounterforce}(X, p, \alpha), \\ &\Rightarrow \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha), \\ &\Rightarrow \text{OfficialCounterexampleForce}_R(F_X(\alpha) = \emptyset), \\ &\Rightarrow \text{EndpointUse}_R(F_X(\alpha) = \emptyset), \\ &\Rightarrow K_R \Rightarrow A_R^+. \end{aligned}$$

Under the official Hodge route, empty-fiber endpoint counterforce is native empty-fiber counterforce. The native route is exhausted:

$$\begin{aligned} \text{NativeEmptyFiberCounterforce}(X, p, \alpha) &\Rightarrow \text{BridgeNeutralized} \vee \text{SupportOnly} \vee \text{Bookkeeping} \\ &\vee \text{DomainShift} \vee D_{\text{fib}}(X, p, \alpha). \end{aligned}$$

In the live subproof, bridge-neutralization contradicts the live empty-fiber assumption; support-only and bookkeeping lack endpoint counterforce; domain shift is not the fixed Hodge endpoint. Hence $D_{\text{fib}}(X, p, \alpha)$. But endpoint use gives $N_{\text{fib}}(X, p, \alpha)$, and by definition

$$N_{\text{fib}}(X, p, \alpha) \Rightarrow \neg D_{\text{fib}}(X, p, \alpha).$$

So the exact-complement subproof yields contradiction and discharges the sole object-level assumption $[F_X(\alpha) = \emptyset]_i$. Therefore $F_X(\alpha) \neq \emptyset$. Universal generalization gives the Hodge Conjecture.

3 Official Hodge endpoint and target instantiation

3.1 Classical endpoint retained

Definition 3.1 (Cycle-fiber carrier). *Let X be a smooth projective complex variety and p a codimension. Define $\text{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$, let $\text{cl}_{\mathbb{Q}} : \text{CH}^p(X)_{\mathbb{Q}} \rightarrow H^{2p}(X, \mathbb{Q})$ be the rational cycle-class map, and let $F_X(\alpha) = \{Z \in \text{CH}^p(X)_{\mathbb{Q}} : \text{cl}_{\mathbb{Q}}(Z) = \alpha\}$.*

Definition 3.2 (Positive and negative branch slots). *The positive local branch is $F_X(\alpha) \neq \emptyset$. The bare local negative branch is $F_X(\alpha) = \emptyset$. The global endpoint is $\forall X, p, \alpha (\alpha \in \text{Hdg}^p(X) \Rightarrow F_X(\alpha) \neq \emptyset)$.*

Theorem 3.3 (Official endpoint identity). *The universal nonempty cycle-fiber statement is the classical Hodge endpoint in rational cycle-class form.*

Proof. The condition $F_X(\alpha) \neq \emptyset$ is precisely the existence of $Z \in \text{CH}^p(X)_{\mathbb{Q}}$ with $\text{cl}_{\mathbb{Q}}(Z) = \alpha$, equivalently membership of α in the image of the rational cycle-class map. Universalizing over smooth projective complex varieties, codimensions, and rational Hodge classes gives the usual Hodge assertion. $\square$

Audit note. This theorem identifies the endpoint; it does not assert the positive branch.

Definition 3.4 (Official Hodge endpoint context). *$C_{\text{HC}}(R, X, p, \alpha)$ is the neutral context object that records the official Hodge carrier and endpoint slots: the smooth projective complex variety X , codimension p , rational class α , the Hodge-standing predicate, the rational Chow group $\text{CH}^p(X)_{\mathbb{Q}}$, the rational cycle-class map $\text{cl}_{\mathbb{Q}}$, the fiber $F_X(\alpha)$, the positive endpoint slot $F_X(\alpha) \neq \emptyset$, the negative slot $F_X(\alpha) = \emptyset$, the empty-fiber separator slot, and the official Hodge route. It is a carrier-and-slot object, not a truth or endpoint-standing assertion.*

Theorem 3.5 (Context non-smuggling). *The Hodge endpoint context fixes the carrier and branch slots but asserts neither nonempty nor empty fiber, neither A^+ nor D_{fib} .*

Proof. A context object declares what is being evaluated: X, p, α , the Hodge-standing predicate, the Chow carrier, the rational cycle-class map, the fiber, and the two endpoint slots. None of these declarations occupies either slot. Occupation or non-occupation enters only through proof roles. $\square$

Audit note. This is the first anti-circularity lock.

Theorem 3.6 (Target-instantiation theorem). *Under the official Hodge route, algebraic readout is exactly nonempty cycle-fiber occupation, and its exact complement is empty-fiber non-occupation.*

Proof. The official route reads Hodge standing through the rational cycle-class map into the rational Chow fiber. Thus positive endpoint role is $F_X(\alpha) \neq \emptyset$, and the exact complement is $F_X(\alpha) = \emptyset$. This fixes roles, not truth. $\square$

Audit note. This theorem is the Hodge analogue of endpoint-image projection.

4 Cycle-class carrier and model adequacy

Lemma 4.1 (Smooth projective carrier exactness). *The carrier is the classical smooth projective complex variety carrier.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.2 (Hodge-standing exactness). *StandHdg(X, p, α) is exactly rational (p, p) classhood.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.3 (Chow carrier exactness). *The readout carrier is $\text{CH}^p(X)_{\mathbb{Q}}$, not a surrogate unless a bridge resets the target.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.4 (Cycle-class exactness). *The readout map is the rational cycle-class map $\text{cl}_{\mathbb{Q}}$.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.5 (Fiber exactness). *$F_X(\alpha)$ is exactly the set of rational Chow classes mapping to α .*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.6 (Readout exactness). *$\text{ReadAlg}(X, p, \alpha) \leftrightarrow F_X(\alpha) \neq \emptyset$.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.7 (Empty-fiber exactness). *$\neg \text{ReadAlg}(X, p, \alpha) \leftrightarrow F_X(\alpha) = \emptyset$.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.8 (No algebraicity smuggling). *The carrier does not include nonempty-fiber occupation by definition.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Lemma 4.9 (Surrogate separation). *Motives, absolute Hodge classes, Hodge loci, Abel-Jacobi data, and known cases are support or bridge programs until they supply the Chow fiber.*

Proof. This is a model-adequacy clause. It records the official carrier and the corresponding readout role. It supplies no positive endpoint occupant. $\square$

Audit note. If denied, the referee must provide a different official Hodge carrier without changing the problem.

Theorem 4.10 (Model adequacy assembly). *The neutral Hodge context and fixed-domain preservation imply the context-bound Hodge/cycle model adequacy packet.*

Proof. The context fixes X, p, α , the Hodge-standing predicate, the Chow group, the rational cycle-class map, and the cycle fiber. Fixed-domain preservation prevents silent changes of coefficients, equivalence relation, target class, or readout slot. Thus the adequacy clauses are in force. $\square$

Audit note. This supplies target instantiation, not a cycle construction.

5 Kernel, A+, ATS, and UEAP replay for Hodge endpoint use

Theorem 5.1 (Target adequacy forces the kernel roles). *Target determinacy, step evaluability, act-time finality, and same-regime fidelity force reference, standing, admissibility, and irreversibility for Hodge endpoint acts.*

Proof. Reference fixes the object of evaluation; standing fixes endpoint usability; admissibility fixes the boundary between licensed continuation and carrier shift; irreversibility blocks later repair as the same act. Without these roles, the alleged Hodge endpoint act becomes raw notation or a different target. $\square$

Audit note. This is the local kernel replay.

Theorem 5.2 (Endpoint use forces A+). *Endpoint use on the fixed Hodge/cycle carrier triggers the A+ consequence layer.*

Proof. Endpoint use is governed construction. The fixed-domain kernel consequences supply AMetric no-selector discipline, no carrier transfer, no repair, report preservation, unique admissible interior, standing conservation, and no independent same-domain endpoint discriminator. $\square$

Audit note. A+ is downstream consequence machinery.

Theorem 5.3 (ATS locking). *An endpoint act preserving the Hodge route must preserve X, p, α , Hodge standing, Chow carrier, cycle-class map, and the fiber slot.*

Proof. ATS separates anchor, tensor, and skin. Presentation may vary, but the anchor and tensor data cannot vary while preserving the same endpoint act. A surrogate readout changes tensor, not skin. $\square$

Audit note. This blocks hidden carrier drift.

Theorem 5.4 (UEAP no-overreporting). *A Hodge-standing, surrogate, obstruction, or support report cannot be issued as endpoint readout or endpoint defeat without the corresponding bridge or endpoint-use role.*

Proof. UEAP preserves target, carrier, dependencies, blockers, and report level. Lower-status material remains lawful but cannot be reported as higher endpoint status by wording alone. $\square$

Audit note. This protects both positive and negative support content.

6 Counterexample force as endpoint use

This section supplies the front-loaded dependency in theorem form. The distinction is not optional: the manuscript does not classify an object, formula, or theorem as endpoint use merely because it is negative. It classifies the act of using that content to defeat the official endpoint as endpoint use. The distinction between content and use is the hinge that prevents the proof from becoming a schema of the form $\neg P \Rightarrow P$.

Definition 6.1 (Bare empty-fiber content). *For fixed X, p, α , the bare empty-fiber content is the raw mathematical formula*

$$\text{BareEmptyFiberContent}(X, p, \alpha) := (\alpha \in \text{Hdg}^p(X) \wedge F_X(\alpha) = \emptyset).$$

It is content at the bare proposition layer. It is not by itself theoremhood, endpoint use, official counterexample force, reportability, downstream reuse, or branch exclusion.

Definition 6.2 (Ordinary empty-fiber theoremhood).

$$\text{OrdEmptyFiberThm}_R := \text{Proof}_R(\exists X, p, \alpha (\alpha \in \text{Hdg}^p(X) \wedge F_X(\alpha) = \emptyset)).$$

This is ordinary negative theoremhood. It is not the rejected discriminator merely by being negative.

Definition 6.3 (Official empty-fiber counterexample force). $\text{OfficialEmptyFiberCounterexampleForce}_R(X, p, \alpha)$ means that empty-fiber content for the same fixed Hodge/cycle carrier is used to defeat or settle the official Hodge endpoint negatively. The use fixes the official target, the variety X , codimension p , rational class α , the Hodge-standing predicate, $\text{CH}^p(X)_{\mathbb{Q}}$, the rational cycle-class map, the fiber $F_X(\alpha)$, the negative branch, reportability as endpoint-defeating content, downstream reuse as counterexample material, exclusion of the positive branch, act-time finality, and lawful-redescription stability.

Definition 6.4 (Local empty-fiber counterexample force). $\text{LocalEmptyFiberCounterexampleForce}_{\text{HC}}(X, p, \alpha)$ means that the local assumption $[F_X(\alpha) = \emptyset]_i$ has been opened as the live exact-complement countercase against the endpoint target $F_X(\alpha) \neq \emptyset$ for the same fixed Hodge-standing instance. This is local proof-role force, not global negative endpoint resolution.

Theorem 6.5 (Bare content is not endpoint use).

$$\text{BareEmptyFiberContent}(X, p, \alpha) \not\Rightarrow \text{EndpointUse}_R(F_X(\alpha) = \emptyset).$$

Proof. The formula $F_X(\alpha) = \emptyset$ may appear as a hypothesis, as a possible obstruction, as the conclusion of a negative theorem, or as a raw branch in a proof by cases. None of these uses alone fixes the official endpoint role of the negative branch. Endpoint use requires the additional act of offering that branch as endpoint-defeating or endpoint-resolving content on the fixed carrier. Bare content therefore does not supply endpoint use. $\square$

Audit note. This theorem protects the lawful mathematical status of empty-fiber syntax. The proof does not ban the formula $F_X(\alpha) = \emptyset$.

Theorem 6.6 (Ordinary negative theoremhood is kernel-internal but not the discriminator). *If $\text{OrdEmptyFiberThm}_R$ is genuine theoremhood rather than raw trace, then it is a governed derivation and hence kernel-internal. It is not, merely as theoremhood, $D_{\text{fib}}(X, p, \alpha)$.*

Proof. A genuine theorem is a standing-bearing derivation. It therefore already presupposes reference, standing, admissibility, and irreversibility. The sign of the theorem does not make it defective. The discriminator at issue later is not the fact that a theorem is negative; it is the endpoint-governing use of empty-fiber non-occupation to defeat the official Hodge endpoint on the same fixed carrier. $\square$

Audit note. Negative algebraic geometry is preserved. Only official endpoint-defeating use enters the separator route.

Theorem 6.7 (Official counterexample force is endpoint use). *For every fixed endpoint context R and branch B ,*

$$\text{OfficialCounterexampleForce}_R(B) \Rightarrow \text{EndpointUse}_R(B).$$

Proof. Official counterexample force is not the bare existence of an object. It is the mathematical act of using the object, theorem, or branch to defeat the official endpoint. Such an act fixes the target it defeats; otherwise it is not a counterexample to that endpoint. It fixes the carrier; otherwise it is a counterexample to a different problem. It fixes the branch; otherwise there is no determinate negative endpoint role. It fixes exclusion of the positive branch; otherwise it does not defeat the endpoint. It fixes reportability and downstream reuse; otherwise the alleged counterexample is not usable as official endpoint-defeating content. It fixes act-time finality; otherwise later repair or reclassification would be the same act. Finally, it must be stable under lawful redescription of the same carrier; otherwise it is a carrier shift. These are precisely the features of theorem-bearing endpoint use. Therefore official counterexample force is endpoint use. $\square$

Audit note. The theorem classifies the use, not the object. An object satisfying a negative formula is not an act; offering it as official endpoint defeat is an act.

Corollary 6.8 (No autonomous counterexample endpoint role). *There is no branch B such that $\text{OfficialCounterexampleForce}_R(B)$ and $\neg \text{EndpointUse}_R(B)$ both hold:*

$$\neg \exists B (\text{OfficialCounterexampleForce}_R(B) \wedge \neg \text{EndpointUse}_R(B)).$$

Proof. If such a branch existed, Theorem 6.7 would give $\text{EndpointUse}_R(B)$, contradicting $\neg \text{EndpointUse}_R(B)$. $\square$

Audit note. This is the no-fifth-role lock at the general counterexample-force level.

Theorem 6.9 (Official empty-fiber counterexample force is endpoint use).

$$\text{OfficialEmptyFiberCounterexampleForce}_R(X, p, \alpha) \Rightarrow \text{EndpointUse}_R(F_X(\alpha) = \emptyset).$$

Proof. The premise is the Hodge specialization of official counterexample force: the empty-fiber branch is used to defeat the official endpoint for the same X, p, α and the same rational cycle-class map. Applying Theorem 6.7 gives endpoint use of the empty-fiber branch. $\square$

Audit note. This is the Hodge specialization of the front-loaded dependency.

Theorem 6.10 (Local exact-complement counterexample force is local endpoint use). *In the local reductio subproof,*

$$\text{ReductioCounterforce}_{\text{HC}}(F_X(\alpha) = \emptyset; F_X(\alpha) \neq \emptyset) \wedge \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha) \Rightarrow \text{LocalEndpointUse}_R(F_X(\alpha) = \emptyset)$$

Proof. The reductio annotation records that the sole opened assumption is the live exact complement of the target. Endpoint counterforce records that this assumption is being used to block the endpoint closure on the fixed carrier. Together they give the local version of official counterexample force: the branch is endpoint-facing, carrier-preserving, target-fixing, and branch-excluding inside the subproof. It is local endpoint use. The conclusion is local; it does not promote the assumption to global official negative resolution. $\square$

Audit note. Local endpoint use triggers the local A^+ consequence layer without treating a local assumption as a global theorem.

Corollary 6.11 (Local endpoint use triggers local A^+).

$$\text{LocalEndpointUse}_R(F_X(\alpha) = \emptyset) \wedge \text{FixedDomain}(R) \Rightarrow A_{R,\text{loc}}^+(F_X(\alpha) = \emptyset).$$

Proof. Endpoint use is governed construction at the local proof-act level. Fixed-domain closure supplies the same kernel-forced consequences locally: no hidden selector, no second same-domain endpoint classifier, report preservation, and no carrier-shift repair. These consequences do not suspend inside a reductio subproof. $\square$

Audit note. The local contradiction later uses this local A^+ layer, not a global negative endpoint outcome.

7 Structural-to-cycle-fiber equivalence bridge

Theorem 7.1 (Hodge standing fixes the cycle-fiber readout slot). *For a fixed Hodge-standing class α , the algebraic-readout slot is $F_X(\alpha)$.*

Proof. The official Hodge route sends rational Hodge standing through the rational cycle-class map. The target readout is therefore the rational Chow fiber over α . $\square$

Audit note. Slot identification is not slot occupation.

Theorem 7.2 (Algebraic readout equivalence). $\text{ReadAlg}(X, p, \alpha) \leftrightarrow F_X(\alpha) \neq \emptyset$.

Proof. Both sides say that some rational Chow class maps to α under $\text{cl}_{\mathbb{Q}}$. $\square$

Audit note. This is the equivalence bridge.

Theorem 7.3 (Empty-fiber equivalence). $\neg \text{ReadAlg}(X, p, \alpha) \leftrightarrow F_X(\alpha) = \emptyset$.

Proof. Negating the nonempty-fiber condition gives empty fiber. This is the exact local complement used in reductio. $\square$

Audit note. The negative branch is exact complement, not a surrogate.

Theorem 7.4 (Projection-role content). *Under the official Hodge route, the exact empty-fiber counterforce has cycle-fiber non-occupation as its endpoint-role content.*

Proof. The official route fixes the positive endpoint image as fiber occupation. Therefore the live complement is non-occupation of that image. The statement remains non-explosive until endpoint counterforce is added. $\square$

Audit note. This is the bridge from raw negation to endpoint role content.

8 Single cycle-fiber endpoint interface

Theorem 8.1 (Single cycle-fiber interface). *On the fixed Hodge endpoint image, algebraic readout has exactly one occupation interface: $F_X(\alpha) \neq \emptyset$.*

Proof. Algebraic readout is defined by nonempty rational cycle-class fiber. Surrogates may support bridges, but they do not occupy the readout slot without providing an element of that fiber. $\square$

Audit note. This is the single-interface theorem.

Corollary 8.2 (Empty fiber is non-occupation, not coequal occupation). *$F_X(\alpha) = \emptyset$ implies non-occupation of the cycle-fiber endpoint image.*

Proof. If the fiber is empty, the endpoint image $F_X(\alpha) \neq \emptyset$ is not occupied. This is not a second occupant of the same image. $\square$

Audit note. The broad outcome carrier $\{HC, \neg HC\}$ remains legitimate but is not the endpoint-image carrier.

Theorem 8.3 (Empty-fiber non-explosiveness). *$F_X(\alpha) = \emptyset$ does not imply contradiction.*

Proof. Empty-fiber content may be raw, descriptive, obstruction-level, support-level, hypothetical, or ordinary theoremhood. The contradiction arises only after endpoint counterforce and separator occupation are in force. $\square$

Audit note. This protects negative algebraic-geometric content.

9 Native empty-fiber counterforce exhaustion

This is the Hodge-local technical core. The theorem to prove is not that empty fiber is impossible as raw content. It is that empty-fiber content used with endpoint-defeating force on the fixed Hodge/cycle carrier has no autonomous ordinary-negative endpoint role outside the endpoint-role trichotomy. The possible roles are exhausted into bridge-neutralized status, support-only status, bookkeeping, domain shift, or the independent cycle-fiber discriminator.

Definition 9.1 (Native empty-fiber counterforce). *NativeEmptyFiberCounterforce(X, p, α) means that the empty-fiber content $F_X(\alpha) = \emptyset$ is used as endpoint-defeating force while preserving the same smooth projective variety X , codimension p , rational class α , Hodge-standing predicate, rational Chow group, rational cycle-class map, and cycle-fiber readout slot. It is native because it uses the official Hodge/cycle carrier; it is counterforce because it is offered to defeat the positive endpoint.*

Definition 9.2 (Bridge-neutralized route). *BridgeNeutralized_{fb}(X, p, α) means that the apparent empty-fiber counterforce is neutralized by a lawful bridge: a witness $Z \in \text{CHP}(X)_{\mathbb{Q}}$ with $\text{cl}_{\mathbb{Q}}(Z) = \alpha$, or a certificate of nonempty fiber. Bridge-neutralization is compatible with positive endpoint readout and incompatible with live endpoint-defeating empty-fiber force as the same local countercase.*

Definition 9.3 (Support-only and bookkeeping routes). *SupportOnly_{fb}(X, p, α) means the empty-fiber material is hypothetical, obstruction-level, local, or lower-status support and is not used to defeat the official endpoint. Bookkeeping_{fb}(X, p, α) means the empty-fiber label changes no endpoint standing, reportability, downstream reuse, branch exclusion, or admissible continuation.*

Definition 9.4 (Domain-shift route). *DomainShift_{fb}(X, p, α) means that the negative branch changes at least one admissibility-relevant component of the fixed Hodge endpoint: the variety X , codimension p , class α , coefficient field, Chow group, equivalence relation, cycle-class map, cohomology theory, readout carrier, or surrogate category. A domain-shift route may be mathematically useful, but it is not the same fixed endpoint.*

Definition 9.5 (Fiber discriminator). $D_{\text{fib}}(X, p, \alpha)$ is the independent same-domain endpoint-status discriminator induced when empty-fiber non-occupation is used to defeat cycle-fiber endpoint occupation on the same fixed carrier while not being bridge-neutralized, support-only, bookkeeping, or domain-shifting.

9.1 The no-sixth-role closure for ordinary native empty-fiber falsification

The previous definitions identify the four harmless or exiting roles and the discriminator role. The hostile objection that remains after those definitions is sharper: perhaps there is an additional role called ordinary native falsification. This subsection proves that there is no such sixth role. The proof is not a dismissal of native algebraic geometry. It is a role-exhaustion theorem for a specific use: same-carrier empty-fiber content with endpoint-defeating force.

Definition 9.6 (Endpoint-impact bundle). For the fixed Hodge endpoint instance (R, X, p, α) , let

$$\mathcal{E}_{\text{HC}}(X, p, \alpha)$$

denote the endpoint-impact bundle consisting of the following features: target fixation, carrier fixation, branch fixation, endpoint standing, endpoint reportability, downstream reuse, exclusion of the positive fiber-occupation branch, act-time finality, and lawful-redescription stability. A use of empty-fiber content is endpoint-impacting when changing or removing that use changes at least one component of this bundle.

Definition 9.7 (Autonomous ordinary-negative endpoint role). An autonomous ordinary-negative endpoint role for (X, p, α) is a use of $F_X(\alpha) = \emptyset$ satisfying all of the following:

1. it is native to the declared Hodge/cycle vocabulary;
2. it preserves the same X, p, α , Chow carrier, rational cycle-class map, and fiber slot;
3. it has endpoint-defeating force against $F_X(\alpha) \neq \emptyset$;
4. it is not bridge-neutralized, support-only, bookkeeping, or domain-shifting;
5. it is asserted not to be $D_{\text{fib}}(X, p, \alpha)$.

This definition isolates the hostile-referee sixth role. It is not assumed to be impossible by definition; it is excluded below.

Lemma 9.8 (Endpoint impact versus bookkeeping). If a use of $F_X(\alpha) = \emptyset$ changes no component of $\mathcal{E}_{\text{HC}}(X, p, \alpha)$, then it is bookkeeping for endpoint purposes.

Proof. The endpoint-impact bundle lists exactly the features by which a fixed endpoint act differs from inert description: target, carrier, branch, standing, reportability, reuse, exclusion, act-time status, and lawful-redescription behavior. If a use of empty-fiber content changes none of these, then it does no endpoint work. A role that does no endpoint work may still be useful notation, a reminder, a syntactic label, or local organization, but it is bookkeeping with respect to the endpoint. Therefore any non-bookkeeping empty-fiber role must change at least one component of the bundle. $\square$

Audit note. This lemma prevents a hidden sixth role from doing endpoint work while being described as inert ordinary language.

Lemma 9.9 (Endpoint impact plus carrier change is domain shift). If endpoint-impacting empty-fiber use changes X, p, α , the coefficient field, the Chow group, the rational equivalence relation, the cycle-class map, the fiber slot, or the declared Hodge/cycle readout carrier, then it is $\text{DomainShift}_{\text{fib}}(X, p, \alpha)$.

Proof. The fixed Hodge endpoint is individuated by these data. A change in any one of them changes the object whose fiber is being evaluated or the map by which readout is typed. The resulting use may be a lawful comparison, a bridge program, or a new endpoint. It is not the same endpoint act. Thus endpoint-impacting use that changes the invariant bundle is domain shift. $\square$

Audit note. Motives, absolute Hodge classes, and alternative equivalence relations are protected as separate carriers, not silently identified with the Chow-fiber endpoint.

Lemma 9.10 (Endpoint impact without carrier change is endpoint-status work). *Assume a use of $F_X(\alpha) = \emptyset$ is endpoint-impacting and preserves the fixed Hodge/cycle carrier. Then it performs endpoint-status work unless it is support-only or bridge-neutralized.*

Proof. Because the carrier is preserved, the use is not merely a neighboring problem. Because it is endpoint-impacting, it changes some member of the endpoint-impact bundle. If the change does not affect branch standing, reportability, reuse, exclusion, or admissible continuation, then by the previous bookkeeping lemma it would be endpoint-inert, contrary to endpoint impact. Hence it affects the status of the endpoint act. If the use remains lower-level obstruction or local evidence, it is support-only and does not have endpoint force. If a lawful bridge supplies nonempty fiber or neutralizes the asserted non-occupation, it is bridge-neutralized. Outside those two cases, the use is endpoint-status work on the same carrier. $\square$

Audit note. This lemma is the role-theoretic bridge from “ordinary falsification used at endpoint level” to endpoint-status work.

Lemma 9.11 (Endpoint-status work by empty fiber is not positive occupation). *Same-carrier endpoint-status work by $F_X(\alpha) = \emptyset$ is not governance-equivalent to positive cycle-fiber occupation unless bridge-neutralized.*

Proof. Positive cycle-fiber occupation is $F_X(\alpha) \neq \emptyset$. Empty-fiber content is $F_X(\alpha) = \emptyset$. On the fixed fiber these are exact complements. If empty-fiber work becomes governance-equivalent to positive occupation, then a bridge has neutralized or overturned the empty-fiber role by supplying a nonempty-fiber certificate. That is the bridge-neutralized branch. If no such bridge is supplied, empty-fiber endpoint-status work excludes rather than occupies the positive endpoint image and is not governance-equivalent to it. $\square$

Audit note. This theorem does not assert nonempty fiber. It states the role difference between occupation and non-occupation on the same fixed fiber.

Lemma 9.12 (Endpoint-status work by empty fiber is not lawful coequal occupation). *Same-carrier endpoint-status work by $F_X(\alpha) = \emptyset$ is not lawful coequal occupation of the cycle-fiber endpoint image.*

Proof. The cycle-fiber endpoint image has one occupation interface: nonempty fiber. A broad problem-outcome carrier may record two outcomes, Hodge and non-Hodge. That broad outcome carrier is not the endpoint image carrier whose occupant is an element or nonempty certificate of $F_X(\alpha)$. Empty fiber supplies non-occupation of that image. Treating non-occupation as coequal occupation changes the interface from cycle-fiber occupation to a symmetric outcome ledger. That is an interface shift, not a same-image occupation. $\square$

Audit note. This is the Hodge counterpart of the SAT/RH single-interface discipline.

Lemma 9.13 (Endpoint-status work not equivalent, not coequal, not shifting is independent). *If empty-fiber endpoint-status work preserves the fixed carrier, is not bridge-neutralized, is not support-only, is not bookkeeping, is not domain-shifting, is not governance-equivalent to positive occupation, and is not lawful coequal occupation of the same endpoint image, then it is $D_{\text{fib}}(X, p, \alpha)$.*

Proof. The endpoint-discriminator trichotomy for a fixed endpoint carrier has four possibilities for a status-determining role: bookkeeping, governance-equivalence to the admitted interface, domain shift, or independent same-domain endpoint-status work. The hypotheses exclude bookkeeping and domain shift. The previous two lemmas exclude governance-equivalence to positive occupation and lawful coequal image occupation. The role remains endpoint-status work on the same carrier. Therefore the only remaining trichotomy branch is independent same-domain endpoint-status discrimination. By definition this is $D_{\text{fib}}(X, p, \alpha)$. $\square$

Audit note. This is where “ordinary native falsification” is forced to declare its role. If it is endpoint-status work and not one of the harmless/exiting roles, it is the discriminator.

Theorem 9.14 (No autonomous ordinary-negative endpoint role). *For the fixed Hodge/cycle endpoint, there is no autonomous ordinary-negative endpoint role for empty-fiber counterforce. Equivalently, any native same-carrier use of $F_X(\alpha) = \emptyset$ with endpoint-defeating force is bridge-neutralized, support-only, bookkeeping, domain-shifting, or $D_{\text{fib}}(X, p, \alpha)$.*

Proof. Let a native same-carrier empty-fiber use have endpoint-defeating force. If a lawful bridge supplies or restores nonempty-fiber occupation, the use is bridge-neutralized. If the use does not actually carry endpoint defeat but only evidential, hypothetical, obstruction, or local analytic content, it is support-only. If it changes no component of the endpoint-impact bundle, it is bookkeeping. If it changes any component of the fixed carrier or readout invariant bundle, it is domain shift. Suppose none of these alternatives holds. Then the use is endpoint-impacting, fixed-carrier, endpoint-status work. It is not governance-equivalent to positive occupation unless bridge-neutralized, already excluded. It is not lawful coequal occupation of the same cycle-fiber endpoint image, because empty fiber is non-occupation of that image. The endpoint-discriminator trichotomy therefore leaves independent same-domain endpoint-status work, namely $D_{\text{fib}}(X, p, \alpha)$. Thus the alleged autonomous ordinary-negative role collapses into one of the five listed roles after all. $\square$

Audit note. This is the main technical closure. A critic must now produce a same-carrier endpoint-defeating empty-fiber role that escapes all five branches; merely naming it “ordinary falsification” does not add a sixth role.

Corollary 9.15 (Ordinary native falsification is status-disambiguated). *The phrase “ordinary native empty-fiber falsification” is not a primitive sixth endpoint role. If it has no endpoint-defeating force, it is raw content, theoremhood, support, or bookkeeping. If it has endpoint-defeating force on the fixed Hodge/cycle carrier, Theorem 9.14 places it in the five-role exhaustion.*

Proof. If the use lacks endpoint-defeating force, it does not settle or oppose the endpoint and therefore falls below endpoint status. If it has endpoint-defeating force, it is a native same-carrier empty-fiber endpoint use and the no-autonomous-role theorem applies. Hence the phrase does not name a sixth endpoint role. $\square$

Audit note. This closes the hostile objection that empty fiber is simply native falsification outside AASC governance.

Lemma 9.16 (Bridge-neutralized force is incompatible with the live empty-fiber counterforce). *Inside the live exact-complement subproof,*

$$F_X(\alpha) = \emptyset \wedge \text{BridgeNeutralized}_{\text{fib}}(X, p, \alpha) \Rightarrow \perp.$$

Proof. Bridge-neutralization supplies the lawful positive bridge: a nonempty fiber witness or certificate. The live counterforce asserts that the same fixed fiber is empty. Because the carrier, class, cycle-class map, and fiber slot are fixed, these cannot both be the same local endpoint branch. If the bridge is admitted, the live empty-fiber counterforce has been neutralized and is no longer the opened exact-complement counterforce. $\square$

Audit note. The contradiction is local to the fixed fiber. It does not say that bridge programs are illicit.

Lemma 9.17 (Support-only lacks endpoint counterforce).

$$\text{SupportOnly}_{\text{fib}}(X, p, \alpha) \Rightarrow \neg \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha).$$

Proof. Support-only material may motivate, obstruct, suggest, or locally analyze the endpoint. By definition it is not used to defeat the official endpoint. Endpoint counterforce is the role of blocking or displacing the endpoint closure. The two roles therefore conflict. $\square$

Audit note. Empty-fiber support content is lawful, but it does not close the endpoint negatively.

Lemma 9.18 (Bookkeeping lacks endpoint counterforce).

$$\text{Bookkeeping}_{\text{fib}}(X, p, \alpha) \Rightarrow \neg \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha).$$

Proof. Bookkeeping affects no endpoint standing, reportability, downstream reuse, branch exclusion, or admissible continuation. Endpoint counterforce affects the endpoint by opposing the positive branch. A label that affects none of those features cannot at the same time be endpoint counterforce. $\square$

Audit note. This prevents hidden fifth-role labels from doing endpoint work without admitting it.

Lemma 9.19 (Domain shift leaves the fixed endpoint).

$$\text{DomainShift}_{\text{fib}}(X, p, \alpha) \Rightarrow \neg \text{FixedDomain}_{\text{HC}}(R, X, p, \alpha).$$

Proof. The fixed Hodge endpoint is individuated by X, p, α , the rational Hodge-standing predicate, $\text{CH}^p(X)_{\mathbb{Q}}$, the rational cycle-class map, and the fiber $F_X(\alpha)$. Changing any of these changes the admissibility-relevant invariant bundle. Such a route may define a new problem or a surrogate bridge program, but it is not the same endpoint act. $\square$

Audit note. This protects motives, absolute Hodge structures, and other frameworks as bridge programs while denying them silent endpoint identity.

Lemma 9.20 (Endpoint-governing empty-fiber work is discriminator work). *Suppose empty-fiber content is endpoint-governing on the fixed Hodge/cycle carrier and is not bridge-neutralized, not support-only, not bookkeeping, and not a domain shift. Then it induces $D_{\text{fib}}(X, p, \alpha)$.*

Proof. Endpoint-governing empty-fiber work determines that the positive cycle-fiber endpoint image is not occupied. Because the carrier is fixed, it is not a different problem. Because it is not support-only or bookkeeping, it changes endpoint standing, reportability, downstream reuse, or branch exclusion. Because it is not bridge-neutralized, it does not collapse into positive fiber occupation. Because it is not domain shift, it remains on the same Hodge/cycle endpoint image. On that image the unique positive occupation interface is $F_X(\alpha) \neq \emptyset$. Empty-fiber endpoint force is therefore a same-domain status determination that is not governance-equivalent to positive occupation. By the endpoint-discriminator trichotomy, the remaining role is independent same-domain endpoint-status discrimination, namely $D_{\text{fib}}(X, p, \alpha)$. $\square$

Audit note. This is the place where ordinary native falsification is collapsed into the endpoint-status trichotomy. A critic must produce a sixth role to avoid the conclusion.

Theorem 9.21 (Native empty-fiber counterforce exhaustion). *For the fixed Hodge/cycle endpoint,*

$$\begin{aligned} \text{NativeEmptyFiberCounterforce}(X, p, \alpha) \Rightarrow & \text{BridgeNeutralized}_{\text{fib}} \vee \text{SupportOnly}_{\text{fib}} \vee \text{Bookkeeping}_{\text{fib}} \\ & \vee \text{DomainShift}_{\text{fib}} \vee D_{\text{fib}}(X, p, \alpha). \end{aligned}$$

Proof. Take native empty-fiber counterforce. If a lawful bridge neutralizes it, the first branch holds. If it has no endpoint-defeating role and merely supplies obstruction, hypothesis, or local material, it is support-only. If it changes no endpoint-relevant feature, it is bookkeeping. If it avoids the fixed fiber by changing the variety, class, coefficient field, equivalence relation, Chow carrier, cycle-class map, or readout carrier, it is domain shift. If none of these four applies, then the empty-fiber content remains same-domain, endpoint-governing, non-bridge-neutralized, non-support, and non-bookkeeping. Lemma 9.20 gives $D_{\text{fib}}(X, p, \alpha)$. $\square$

Audit note. This is the no-autonomous-native-falsification theorem for Hodge.

Theorem 9.22 (Native empty-fiber counterforce collapse). *In the live exact-complement fixed-domain subproof,*

$$F_X(\alpha) = \emptyset \wedge \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha) \wedge \text{FixedDomain}_{\text{HC}}(R, X, p, \alpha) \Rightarrow D_{\text{fib}}(X, p, \alpha).$$

Proof. Endpoint counterforce on the fixed Hodge/cycle carrier gives native empty-fiber counterforce. Apply Theorem 9.21. The bridge-neutralized branch is incompatible with the live empty-fiber assumption by Lemma 9.16. The support-only branch is incompatible with endpoint counterforce by Lemma 9.17. The bookkeeping branch is incompatible with endpoint counterforce by Lemma 9.18. The domain-shift branch is incompatible with fixed-domain evaluation by Lemma 9.19. The only remaining branch is $D_{\text{fib}}(X, p, \alpha)$. $\square$

Audit note. This is the technical closure of the ordinary-negative fifth-role objection.

Corollary 9.23 (No ordinary-negative fifth role). *There is no same-carrier endpoint-defeating empty-fiber role that is not bridge-neutralized, support-only, bookkeeping, domain shift, or $D_{\text{fib}}(X, p, \alpha)$.*

Proof. Such a role would be native empty-fiber counterforce but would avoid every branch in Theorem 9.21. That is impossible. $\square$

Audit note. The phrase “ordinary native falsification” names a use. If the use defeats the endpoint, it is in the exhaustion; if it does not, it is support or bookkeeping.

Main-body closure of the ordinary-native-falsification objection

The prior corollary is the compact closure. This subsection expands it because this is the precise hostile-referee pressure point. The objector’s proposed sixth role is not a motive, a selector, a repair channel, or a known-case transfer. It is the claim that the native formula

$$F_X(\alpha) = \emptyset$$

should be allowed to stand as ordinary falsification of the positive endpoint without becoming endpoint-use discipline or independent discriminator work. The proof below separates content from use and then exhausts the use.

Definition 9.24 (Native falsification package). *A native falsification package for (X, p, α) is a proposed endpoint role $\mathcal{O}_{\text{nat}}(X, p, \alpha)$ carrying the same X , the same p , the same rational class α , the same Chow group $\text{CH}^p(X)_{\mathbb{Q}}$, the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$, the same fiber $F_X(\alpha)$, the negative content $F_X(\alpha) = \emptyset$, and claimed endpoint-defeating force against $F_X(\alpha) \neq \emptyset$.*

Lemma 9.25 (Native falsification below endpoint use). *A native falsification package without claimed endpoint-defeating force is below endpoint use. It is raw content, ordinary theoremhood, obstruction information, support, or bookkeeping.*

Proof. Without endpoint-defeating force, the package does not exclude the positive endpoint branch, does not settle the local endpoint, and does not license downstream reuse as a counterexample. It may remain mathematically meaningful: it may be a formula, a theorem, an obstruction statement, a local hypothesis, or a support datum. But it does not yet have the role of defeating the official Hodge endpoint. Therefore it remains below endpoint use. $\square$

Lemma 9.26 (Deployment fixes official counterexample force). *If a native falsification package is deployed to defeat $F_X(\alpha) \neq \emptyset$ on the fixed Hodge/cycle carrier, then it is official empty-fiber counterexample force.*

Proof. Deployment to defeat the endpoint means that the package is not merely recorded but used against the official positive branch. Because the package preserves the same X, p, α , the same Chow group, the same cycle-class map, and the same fiber, the counterexample force is same-carrier force. Because the negative content is $F_X(\alpha) = \emptyset$, the branch is the exact endpoint complement. Thus the deployment is official empty-fiber counterexample force. $\square$

Lemma 9.27 (Deployed native falsification is endpoint use). *A deployed native falsification package is endpoint use.*

Proof. By the previous lemma, a deployed native falsification package is official empty-fiber counterexample force. Official counterexample force is endpoint use by Theorem 6.7. Therefore the deployed native falsification package is endpoint use. $\square$

Lemma 9.28 (Endpoint-use native falsification enters the endpoint-status trichotomy). *A deployed native falsification package that is endpoint use is bookkeeping, governance-equivalent to the positive endpoint interface, independent same-domain endpoint-status work, or a domain shift.*

Proof. Endpoint use fixes the carrier and endpoint role. A status factor attached to that use either changes no endpoint-governance verdict, in which case it is bookkeeping; reproduces the already admitted positive interface, in which case it is governance-equivalent; changes endpoint status on the same carrier without being equivalent, in which case it is independent same-domain endpoint-status work; or changes the admissibility-relevant carrier, in which case it is domain shift. These cases exhaust whether the alleged role changes endpoint work and whether it remains on the fixed carrier. $\square$

Lemma 9.29 (Deployed empty-fiber falsification is not governance-equivalent to positive occupation). *A deployed native empty-fiber falsification package cannot be governance-equivalent to the positive cycle-fiber endpoint interface while preserving endpoint-defeating negative force.*

Proof. The positive endpoint interface is $F_X(\alpha) \neq \emptyset$. The deployed native falsification package asserts and uses $F_X(\alpha) = \emptyset$ to defeat that interface. If it were governance-equivalent to positive occupation, it would not exclude the positive endpoint branch. If it excludes the positive endpoint branch, it performs a different endpoint-status role. Hence, while retaining endpoint-defeating force, it is not governance-equivalent to positive fiber occupation. $\square$

Theorem 9.30 (Ordinary native falsification collapse, expanded form). *A native falsification package for (X, p, α) has exactly the following same-mode possibilities: it is raw, support, or theorem-level content below endpoint use; it is bridge-neutralized; it is bookkeeping; it is a domain shift; or it is the independent cycle-fiber endpoint discriminator $D_{\text{fb}}(X, p, \alpha)$. There is no autonomous ordinary-negative endpoint role outside this list.*

Proof. If the package is not deployed to defeat the endpoint, it is below endpoint use by the first lemma. If a lawful bridge neutralizes it, it is bridge-neutralized. If it changes no endpoint-relevant feature, it is bookkeeping. If it changes the carrier, class, Chow group, cycle-class map, coefficient field, equivalence relation, or readout slot, it is domain shift. It remains to consider the package deployed to defeat the endpoint while preserving the fixed carrier and while not being bridge-neutralized or bookkeeping. By the deployment lemma and the endpoint-use lemma, it is endpoint use. By the endpoint-status trichotomy, it must be bookkeeping, governance-equivalent, independent same-domain work, or domain shift. Bookkeeping and domain shift have already been removed. Governance-equivalence to the positive endpoint is impossible by the previous lemma. Therefore the remaining case is independent same-domain endpoint-status work. In the fixed Hodge/cycle endpoint image, such work induced by empty-fiber non-occupation is precisely $D_{\text{fb}}(X, p, \alpha)$. $\square$

Corollary 9.31 (No sixth role in the live subproof, expanded form). *In the exact-complement reductio subproof, the opened assumption $[F_X(\alpha) = \emptyset]_i$ cannot be deployed as ordinary native endpoint falsification without yielding $D_{\text{fib}}(X, p, \alpha)$.*

Proof. The opened assumption is used as the live endpoint counterforce against $F_X(\alpha) \neq \emptyset$. Thus it is deployed to defeat the endpoint, not merely recorded as raw syntax. The fixed-domain hypotheses remove domain shift. Endpoint counterforce removes support-only and bookkeeping readings. The live empty-fiber assumption removes bridge-neutralized positive occupation. By the expanded collapse theorem, the remaining role is $D_{\text{fib}}(X, p, \alpha)$. $\square$

Technical closure achieved. After this subsection, the objection “empty fiber is just ordinary native falsification” is not an unaddressed case. It is a challenge to a named theorem: deployment fixes official counterexample force; official counterexample force is endpoint use; endpoint-use native falsification enters the trichotomy; and a deployed empty-fiber branch is not governance-equivalent to positive fiber occupation.

10 Cycle-fiber separator and discriminator closure

Definition 10.1 (Local empty-fiber separator occupation). *EmptyFibSep^{loc}(X, p, α) means that cycle-fiber non-occupation is used inside the exact-complement reductio subproof as the endpoint-defeating negative branch on the fixed Hodge/cycle carrier.*

Definition 10.2 (Global empty-fiber separator occupation). *EmptyFibSep^{occ}(X, p, α) means that cycle-fiber non-occupation is used as the official negative endpoint resolution on the fixed Hodge/cycle carrier.*

Theorem 10.3 (Endpoint counterforce to local separator).

$$\text{ProjectionRoleContent}_{\text{cyc}}(F_X(\alpha) = \emptyset, X, p, \alpha) \wedge \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha) \Rightarrow \text{EmptyFibSep}^{\text{loc}}(X, p, \alpha).$$

Proof. Projection-role content binds the local negative branch to non-occupation of the cycle-fiber endpoint image. Endpoint counterforce says this non-occupation is being used to defeat the positive endpoint inside the subproof. That is precisely local empty-fiber separator occupation. $\square$

Audit note. Raw non-occupation alone is not separator occupation; endpoint counterforce is required.

Theorem 10.4 (Local separator induces the fiber discriminator).

$$\text{EmptyFibSep}^{\text{loc}}(X, p, \alpha) \Rightarrow D_{\text{fib}}(X, p, \alpha).$$

Proof. Local separator occupation is endpoint-defeating empty-fiber use on the fixed carrier. Therefore the hypotheses of Theorem 9.22 are met inside the local subproof, and $D_{\text{fib}}(X, p, \alpha)$ follows. $\square$

Audit note. The discriminator is not a descriptive label; it is endpoint-status work.

Theorem 10.5 (Endpoint use triggers no-independent fiber closure).

$$\text{LocalEndpointUse}_R(F_X(\alpha) = \emptyset) \wedge A_{R, \text{loc}}^+ \Rightarrow N_{\text{fib}}(X, p, \alpha).$$

Proof. Local endpoint use instantiates the fixed-domain kernel, and the local A^+ package includes no independent same-domain endpoint-status discriminator. The Hodge specialization of that closure is $N_{\text{fib}}(X, p, \alpha)$. $\square$

Audit note. The no-independent closure binds local counterforce use as well as global official resolution.

Theorem 10.6 (No-independent closure excludes the fiber discriminator).

$$N_{\text{fib}}(X, p, \alpha) \Rightarrow \neg D_{\text{fib}}(X, p, \alpha).$$

Proof. N_{fib} is defined as the absence of independent same-domain cycle-fiber endpoint-status discrimination. D_{fib} is exactly such a discriminator. Therefore N_{fib} excludes it. $\square$

Audit note. This is the Hodge specialization of the A^+ no-independent-discriminator closure.

Corollary 10.7 (Local empty-fiber separator impossible).

$$\text{LocalEndpointUse}_R(F_X(\alpha) = \emptyset) \wedge A_{R,\text{loc}}^+ \Rightarrow \neg \text{EmptyFibSep}^{\text{loc}}(X, p, \alpha).$$

Proof. Assume local separator occupation. Theorem 10.4 gives $D_{\text{fib}}(X, p, \alpha)$. Theorem 10.5 gives $N_{\text{fib}}(X, p, \alpha)$, and Theorem 10.6 gives $\neg D_{\text{fib}}(X, p, \alpha)$. Contradiction. $\square$

Audit note. Empty-fiber non-occupation remains lawful as content; only local separator occupation is impossible.

11 Local reductio and universal Hodge endpoint

The final endpoint step is local. It does not assume a global negative endpoint outcome and does not infer endpoint standing from raw truth. It opens the exact complement of the target for a fixed Hodge-standing instance, routes that local counterexample through endpoint-counterforce discipline, obtains the local separator contradiction, and discharges the original assumption.

Definition 11.1 (Exact-complement reductio annotation). $\text{ReductioCounterCase}_{\text{HC}}(F_X(\alpha) = \emptyset; F_X(\alpha) \neq \emptyset)$ is the proof-act annotation generated when the sole local assumption $[F_X(\alpha) = \emptyset]_i$ is opened as the live exact complement of the target $F_X(\alpha) \neq \emptyset$ for the fixed Hodge-standing instance. The annotation is not a second object-level hypothesis.

Theorem 11.2 (Annotation is not a premise). *The exact-complement reductio annotation is a proof-act role, not an additional object-level assumption.*

Proof. The only object-level formula opened in the subproof is $F_X(\alpha) = \emptyset$. The annotation records the role of that assumption: it is the live exact complement of the target. The later contradiction therefore discharges the original assumption, not a strengthened conjunction. $\square$

Audit note. This blocks the objection that the proof proves only $\neg(F_X(\alpha) = \emptyset \wedge \text{annotation})$.

Theorem 11.3 (Exact-complement assumption has local counterexample standing).

$$\text{ReductioCounterCase}_{\text{HC}}(F_X(\alpha) = \emptyset; F_X(\alpha) \neq \emptyset) \Rightarrow \text{LocalEmptyFiberCounterCase}(X, p, \alpha).$$

Proof. In the fixed endpoint proof, the complement of nonempty fiber is empty fiber. Opening that complement as the live branch whose survival would block the target gives it local counterexample standing. This is temporary derivational standing inside the subproof, not theoremhood of the negative branch. $\square$

Audit note. Inert assumptions remain inert; only the exact-complement endpoint counterexample receives this local role.

Theorem 11.4 (Local counterexample has endpoint counterforce).

$$\text{LocalEmptyFiberCounterCase}(X, p, \alpha) \Rightarrow \text{EndpointCounterforce}_{\text{HC}}(X, p, \alpha).$$

Proof. The local counterexample is introduced precisely to defeat the target $F_X(\alpha) \neq \emptyset$ if it survives. That endpoint-facing role is endpoint counterforce. No global official negative endpoint outcome is inferred. $\square$

Audit note. This is local proof discipline, not a raw-truth-to-standing principle.

Theorem 11.5 (Local empty-fiber counterexample contradiction). *Under fixed Hodge endpoint evaluation, the local assumption $[F_X(\alpha) = \emptyset]_i$ leads to contradiction.*

Proof. Open $[F_X(\alpha) = \emptyset]_i$ as the exact complement of $F_X(\alpha) \neq \emptyset$. By Theorem 11.3, it has local empty-fiber counterexample standing; by Theorem 11.4, it has endpoint counterforce. The official Hodge route gives projection-role content for cycle-fiber non-occupation. Theorem 10.3 gives local separator occupation. Theorem 10.4 gives $D_{\text{fib}}(X, p, \alpha)$. Local endpoint use and the local A^+ package give $N_{\text{fib}}(X, p, \alpha)$ by Theorem 10.5; Theorem 10.6 gives $\neg D_{\text{fib}}(X, p, \alpha)$. Thus the subproof contains contradiction. $\square$

Audit note. The contradiction is local and discharges the exact-complement assumption; it does not promote the assumption into a global endpoint outcome.

Theorem 11.6 (Annotation discharge). *If the annotated exact-complement subproof from $[F_X(\alpha) = \emptyset]_i$ yields contradiction, then the discharged result is $F_X(\alpha) \neq \emptyset$.*

Proof. By Theorem 11.2, the annotation is not an additional object-level assumption. The sole discharged object-level assumption is $F_X(\alpha) = \emptyset$. Classical reductio therefore yields its negation, i.e. $F_X(\alpha) \neq \emptyset$. $\square$

Audit note. This is the Hodge analogue of the exact-complement annotation-discharge rule.

Theorem 11.7 (Arbitrary fixed-instance closeout). *For arbitrary fixed X, p, α ,*

$$\alpha \in \text{Hdg}^p(X) \Rightarrow F_X(\alpha) \neq \emptyset.$$

Proof. Assume $\alpha \in \text{Hdg}^p(X)$ and work in the official Hodge endpoint context for (X, p, α) . Open the exact complement $[F_X(\alpha) = \emptyset]_i$. Theorem 11.5 gives contradiction in the annotated subproof. Theorem 11.6 discharges the original assumption and yields $F_X(\alpha) \neq \emptyset$. $\square$

Audit note. The proof is by local endpoint-counterexample exclusion, not by analytic cycle construction.

Theorem 11.8 (Universal Hodge endpoint).

$$\forall X, p, \alpha (\alpha \in \text{Hdg}^p(X) \Rightarrow F_X(\alpha) \neq \emptyset).$$

Proof. The fixed-instance closeout was proved for arbitrary smooth projective complex X , codimension p , and rational class α in Hodge standing. Universal generalization gives the displayed theorem, which is the Hodge Conjecture in rational cycle-fiber form. $\square$

Audit note. This is the official endpoint. No context clause asserted it upstream.

12 Ordinary practice, route tests, and incompleteness firewall

The following bridge tests preserve ordinary Hodge-theoretic practice while enforcing the fixed endpoint carrier. Each route is lawful as support, bridge material, or local completion when it actually supplies the rational Chow fiber. It is blocked only when it is overreported as endpoint occupation or endpoint defeat without the declared bridge.

Route	Lawful role	Failure if overreported	AASC disposition
Lefschetz (1,1)	local bridge completion in codimension one	universal transfer beyond scope	preserved as scoped bridge completion
Hard Lefschetz	cohomological tensor structure	cohomological structure reported as Chow fiber	support unless Chow fiber supplied
Hodge loci	standing persistence in families	standing persistence reported as cycle witness	support or bridge program
Intermediate Jacobians	obstruction/refinement carrier	obstruction data reported as positive readout	obstruction support unless fiber supplied
Motives	surrogate/enriched carrier	carrier transfer without Chow bridge	domain shift unless bridged back
Absolute Hodge classes	stronger invariance	stronger standing reported as algebraic readout	support unless Chow fiber supplied
Known special cases	local bridge completion	universal transfer without target gating	scoped support, not universal endpoint transfer
Standard conjectures	bridge program	programmatic expectation reported as completed bridge	support until bridge completion
Deformation and spread	family route	local data treated as global fiber without compatibility	support until compatibility certificate

Theorem 12.1 (Same-domain incompleteness firewall). *Independence, undecidability, stronger-theory, or model-theoretic distinctions affect the endpoint only if they change endpoint standing, reportability, downstream reuse, branch exclusion, or admissible continuation.*

Proof. If no endpoint feature changes, the distinction is endpoint-inert. If an endpoint feature changes, it is gate work and falls under A+ endpoint-discriminator discipline. $\square$

Audit note. This does not deny incompleteness phenomena; it denies a third endpoint role.

13 Exhaustion, weakening resistance, and denial burden

Locus	Hodge form	Disposition
Hodge-standing promotion	$\alpha \in \text{Hdg}^p(X)$ reported as $F_X(\alpha) \neq \emptyset$	UEAP/no-generator
Fiber declaration	Nonempty fiber asserted without bridge	report preservation
Empty-fiber counterforce	Empty fiber used as endpoint defeat	native-counterforce collapse
Motive surrogate	Motivic object replaces Chow fiber	no carrier/silent redescription
Absolute-Hodge surrogate	Stronger invariance replaces cycle witness	surrogate separation
Known-case transfer	Local theorem used universally	no-carrier
Local-to-global assembly	Local cycles treated as global fiber	non-factorizability
Selector/canonical representative	Preferred cycle chosen before fiber fixed	AMetric boundary
Repair	Later cycle repairs earlier report	irreversibility
Hidden fifth negative role	Endpoint-defeating empty fiber not endpoint use	no autonomous counterexample endpoint role

Theorem 13.1 (Weakening resistance). *Any strict same-carrier weakening that permits endpoint-defeating empty-fiber counterforce either lowers it to support/bookkeeping, changes carrier, or reintroduces D_{fib} .*

Proof. A weakening that changes no endpoint verdict is bookkeeping. If it has endpoint force on the same carrier, it is endpoint-status work. If it avoids this by changing carrier or readout, it is domain shift. $\square$

Audit note. This prevents a weakened packet from reinstating the negative branch.

13.1 Right-mode objection table

Target	Successful objection must produce
Context non-smuggling	A clause in C_{HC} that asserts nonempty fiber, empty fiber, A^+ , or D_{fib} .
Model adequacy	A mismatch between $\text{ReadAlg}(X, p, \alpha)$ and $F_X(\alpha) \neq \emptyset$, or a different official Hodge carrier.
Counterexample force	A same-carrier official counterexample act that defeats the endpoint while not fixing endpoint-use features.
Cycle-fiber projection	A Hodge endpoint route in which algebraic readout is not cycle-fiber occupation.
Native empty-fiber exhaustion	A fifth role for endpoint-defeating empty-fiber counterforce.
A^+ closure	An independent same-domain cycle-fiber endpoint discriminator compatible with no-independent-discriminator closure.
Local reductio	A proof that the exact-complement annotation is an added object-level premise or discharges the wrong assumption.
Universalization	A failure of arbitrary fixed-instance closure to yield the universal Hodge statement.

14 Conclusion

The manuscript has one nonclassical dependency: official endpoint-defeating counterexample force is theorem-bearing endpoint use. Once that is accepted, the proof is target instantiation plus route exhaustion. The official Hodge endpoint is preserved: every rational Hodge class must have nonempty rational cycle-class fiber. Empty-fiber content is lawful as raw content, descriptive nonmembership, support, obstruction, or ordinary negative theoremhood. It becomes endpoint-governed only when used with official endpoint-defeating force on the same fixed carrier.

The live exact-complement empty-fiber branch cannot occupy an autonomous ordinary-negative endpoint role. Bridge-neutralized, support-only, bookkeeping, and domain-shift routes are unavailable in the local endpoint subproof. The remaining branch is $D_{\text{fib}}(X, p, \alpha)$. Endpoint use supplies $N_{\text{fib}}(X, p, \alpha)$ and hence $\neg D_{\text{fib}}(X, p, \alpha)$. The contradiction discharges the local assumption $F_X(\alpha) = \emptyset$. Thus every rational Hodge class has nonempty rational cycle-class fiber:

$$\forall X, p, \alpha \in \text{Hdg}^p(X), \quad F_X(\alpha) \neq \emptyset.$$

Equivalently, every rational Hodge class is a rational linear combination of classes of algebraic cycles. This is the Hodge Conjecture.

A Theorem ladder and dependency audit

Ladder item	Function
Mathematical-status lock	Proof-order component in the endpoint closeout.
Front-loaded dependency lock	Proof-order component in the endpoint closeout.
Context non-smuggling	Proof-order component in the endpoint closeout.
Model adequacy	Proof-order component in the endpoint closeout.
Kernel forcing	Proof-order component in the endpoint closeout.
Endpoint use forces A+	Proof-order component in the endpoint closeout.
Counterexample force is endpoint use	Proof-order component in the endpoint closeout.
No autonomous counterexample role	Proof-order component in the endpoint closeout.
Cycle-fiber equivalence bridge	Proof-order component in the endpoint closeout.
Single cycle-fiber interface	Proof-order component in the endpoint closeout.
Native empty-fiber exhaustion	Proof-order component in the endpoint closeout.
Separator induction	Proof-order component in the endpoint closeout.
A+ exclusion	Proof-order component in the endpoint closeout.
Local reductio discharge	Proof-order component in the endpoint closeout.
Universal endpoint	Proof-order component in the endpoint closeout.

B Anti-circularity and hostile-misreading audit

Context smuggles nonempty fiber. Context non-smuggling theorem says carrier and slots only.

Empty fiber is illegal. Non-explosiveness preserves empty-fiber content.

Ordinary theoremhood is banned. Ordinary negative theoremhood is protected.

Counterexample object treated as act. Object/use distinction is explicit.

Arbitrary negation becomes separator. Official route, projection content, and endpoint counterforce are required.

Surrogates are dismissed. They remain bridge programs or support.

A+ does algebraic geometry. A+ excludes endpoint discriminator work; it does not construct cycles.

Endpoint changed. The final theorem is universal nonempty rational cycle-class fiber.

C Route-locus exhaustion matrix

This appendix records the route-locus audit without repeating the same closure paragraph for each locus. Every unbridged passage from Hodge standing or empty-fiber content to endpoint readout must alter status, codomain, existence, carrier, time, locality, selector authority, or endpoint-classifier role. The AASC disposition blocks that alteration unless a bridge is supplied or an explicit domain reset is declared.

Locus	What the route changes or attempts	AASC disposition
Status promotion	Hodge standing or obstruction status is reported as endpoint readout	no-generator and UEAP no-overreporting
Codomain substitution	cohomological, motivic, Abel-Jacobi, or variational data replaces the Chow fiber	carrier shift unless a Chow-fiber bridge is supplied
Existence creation	nonempty fiber is asserted without a cycle, class, or bridge certificate	report preservation blocks the promotion

Locus	What the route changes or attempts	AASC disposition
Carrier transfer	coefficients, equivalence relation, cohomology theory, motive, or category changes	explicit domain shift, not the same endpoint
Temporal repair	later cycle discovery is used to repair an earlier unbridged report as the same act	irreversibility: later bridge completion is a new act
Local assembly	local or family cycle data is treated as a global rational Chow fiber	requires compatibility and global-fiber certificate
Selector authority	a preferred representative or canonical choice decides readout before the fiber is fixed	AMetric no-selector boundary
Second classifier	an extra endpoint-status classifier is introduced on the same carrier	bookkeeping, governance-equivalence, domain shift, or forbidden discriminator
Hidden fifth role	endpoint-defeating empty fiber refuses endpoint-use and discriminator classification	no autonomous counterexample endpoint role

D Notation ledger

Symbol	Meaning
X	smooth projective complex variety
p	codimension
α	rational cohomology class
$\text{Hdg}^p(X)$	rational Hodge classes $H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$
$\text{CH}^p(X)_{\mathbb{Q}}$	rational Chow group
$\text{cl}_{\mathbb{Q}}$	rational cycle-class map
$F_X(\alpha)$	cycle-class fiber
ReadAlg	algebraic-readout predicate
D_{fib}	independent cycle-fiber endpoint-status discriminator
N_{fib}	no-independent cycle-fiber closure
$B_X^{\text{cyc}}(\alpha)$	cycle-realization bridge object

E Corpus support ledger

Local role	Source ID	Paper	Object
Kernel	ERK20260527-New_erk_files_May_C3E3B7-LEM-001	A Classification Theorem for Admissi- ble Standard-Model Structures Unique	Envelope necessity
Kernel	ERK20260527-New_erk_files_May_C3E3B7-THM-004	A Classification Theorem for Admissi- ble Standard-Model Structures Unique	Kernel necessity as a derived struc- tural constraint
Kernel	ERK20260527-New_erk_files_May_C3E3B7-COR-002	A Classification Theorem for Admissi- ble Standard-Model Structures Unique	Kernel non- optionality
UEAP	ERK20260527-New_erk_files_May_AF0C6B-DEF-001	Claim Standing and Legitimacy A Uni- versal Audit Protocol for Laundering	ixes an audit ob- ject or status rule

Local role	Source ID	Paper	Object
UEAP	ERK20260527-New_erk_files_May_AF0C6B-PRI-001	Claim Standing and Legitimacy A Universal Audit Protocol for Laundering	Counterexample burden
UEAP	ERK20260527-New_erk_files_May_AF0C6B-REM-002	Claim Standing and Legitimacy A Universal Audit Protocol for Laundering	Why this is the central spine
Claim Standing	ERK20260527-New_erk_files_May_2490B6-PRO-012	Admissibility and the Common Structural Origin of Gauge, Constraint, and	Template character of the fixed-domain closure argument
Claim Standing	ERK20260527-New_erk_files_May_2490B6-THM-019	Admissibility and the Common Structural Origin of Gauge, Constraint, and	Collapse classification under AER-3
Claim Standing	ERK20260527-New_erk_files_May_2490B6-PRO-016	Admissibility and the Common Structural Origin of Gauge, Constraint, and	Conditional summary for unitary multiplicity under AER
ATS	ERK20260527-New_erk_files_May_4C9E73-REM-006	A Fixed-Domain Exhaustion Theorem Standing, Admissibility, and the Determination	Undefinedness is not a third truth value
ATS	ERK20260527-New_erk_files_May_4C9E73-REM-009	A Fixed-Domain Exhaustion Theorem Standing, Admissibility, and the Determination	The axioms below are the explicit premises extracted from the
ATS	ERK20260527-New_erk_files_May_4C9E73-REM-010	A Fixed-Domain Exhaustion Theorem Standing, Admissibility, and the Determination	Detectability is derived, not assumed
AMetric	ERK20260527-New_erk_files_May_C3E3B7-DEF-017	A Classification Theorem for Admissible Standard-Model Structures Unique	AMetric Boundary for admissible bookkeeping
AMetric	ERK20260527-New_erk_files_May_C3E3B7-THM-011	A Classification Theorem for Admissible Standard-Model Structures Unique	AMetric Boundary forcing
AMetric	ERK20260527-New_erk_files_May_C3E3B7-DEF-020	A Classification Theorem for Admissible Standard-Model Structures Unique	Admissibly distinct construction trajectory
No Selection	Post-ERK20260527-New_erk_files_May_311F0C-THM-009	Boundary-Trace Fixation of Continuation Loci under AASC A Role-Anchor Th	No post-selection repair
No Selection	Post-ERK20260527-New_erk_files_May_045695-COR-006	Boundary-Trace Fixation under Admissibility Forced Environmental Coupling	No post-selection environmental repair
No Selection	Post-ERK20260527-New_erk_files_May_D2FE05-COR-002	Constraint Architectures for Coherent Evaluability Quotient Interfaces,	No post-selection boundary repair
Silent description	Re-ERK20260527-New_erk_files_May_FBED12-REM-005	Admissible Construction and the Non-Instantiability of Gödel Architectures	Seminar and blackboard practice

Local role	Source ID	Paper	Object
Silent description	Re- ERK20260527-New_erk_files_May_35B594COR-007	Non-Degenerate Construction and the Kernel of Admissibility An Exhaustio	No silent redescription
Silent description	Re- ERK20260527-New_erk_files_May_9ECD10LEM-011	Rigidity of the Black-Scholes Valuation Operator Representative-Invariant	No silent redescription
Role-Occupancy	ERK20260527-New_erk_files_May_DCCC53REM-001	Role-Occupancy Closure under Admissibility Why role-cardinality matching	The answer is not chosen
Role-Occupancy	ERK20260527-New_erk_files_May_DCCC53DEF-001	Role-Occupancy Closure under Admissibility Why role-cardinality matching	Fixed certificate
Role-Occupancy	ERK20260527-New_erk_files_May_DCCC53DEF-002	Role-Occupancy Closure under Admissibility Why role-cardinality matching	Candidate response class
No generators	ERK20260527-New_erk_files_May_FBED12THM-049	Admissible Construction and the Non-Instantiability of Gödel Architectur	No generators
No generators	ERK20260527-New_erk_files_May_3D011A THM-043	Forty-Two Exhaustion, Closure, and Downstream Consequence	No generators
No generators	ERK20260527-New_erk_files_May_35B594COR-011	Non-Degenerate Construction and the Kernel of Admissibility An Exhaustio	No generators
No carriers	ERK20260527-New_erk_files_May_416051THM-029	The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissib	Admissibility constraint on No carriers (numeric or structur
No carriers	ERK20260527-New_erk_files_May_8B37FDCOR-004	The Bivalence Theorem for Non-Degenerate Construction Standing Conservat	No Carriers
No carriers	ERK20260527-New_erk_files_May_EA5125THM-003	Unus Solus Possibilis Est Corpus Closure and Uniqueness of the Admissibl	No carriers (numeric or structural

F Lean4 Hodge Endpoint Audit Appendix

The manuscript proof above remains the primary proof. The Lean4 material in this appendix is audit support for the local AASC/Hodge endpoint bridge and closeout surface:

$$\begin{aligned}
&\text{OfficialEmptyFiberCounterexampleForce}_R(X, p, \alpha) \Rightarrow \text{EndpointUse}_R(F_X(\alpha) = \emptyset), \\
&\text{EmptyFibSep}(X, p, \alpha) \Rightarrow D_{\text{fib}}(X, p, \alpha), \\
&N_{\text{fib}}(X, p, \alpha) \Rightarrow \neg D_{\text{fib}}(X, p, \alpha), \\
&\text{LocalEmptyFiberCounterexample}(X, p, \alpha) \Rightarrow F_X(\alpha) \neq \emptyset, \\
&\forall X, p, \alpha (\alpha \in \text{Hdg}^p(X) \Rightarrow F_X(\alpha) \neq \emptyset).
\end{aligned}$$

It does not claim to rebuild Hodge theory, Chow groups, algebraic cycles, or the rational cycle-class map from foundations inside Lean. The Hodge/cycle carrier is represented semantically so that the endpoint-use, empty-fiber separator, and no-independent-discriminator route can be checked at the same audit-spine level as the other AASC endpoint archives.

F.1 Release, environment, and audit runner

The Lean audit archive corresponding to this manuscript is:

`somamaley-ux/AASC-Hodge-Endpoint-Lean-Audit`

with public DOI

`10.5281/zenodo.20640238.`

Item	Recorded value
Lean audit repository	<code>https://github.com/somamaley-ux/AASC-Hodge-Endpoint-Lean-Audit</code>
Zenodo DOI	<code>https://doi.org/10.5281/zenodo.20640238</code>
Release	<code>v0.1.0</code>
Lean toolchain	<code>leanprover/lean4:v4.28.0</code>
Pinned mathlib revision	<code>8f9d9cff6bd728b17a24e163c9402775d9e6a365</code>
Main Hodge module	<code>MaleyLean/Papers/Hodge/EndpointClosure.lean</code>
Audit runner target	<code>MaleyLean.Papers.Hodge.AuditRunners</code>
Local audit script	<code>scripts/check-hodge-endpoint-audit.ps1</code>
Focused Hodge audit files	Seven focused files under <code>Checks/Axiom/</code>
Recorded progress	<code>HodgeEndpointClosure=100%</code>
Project-level incomplete declarations	No live project-level <code>axiom</code> , <code>sorry</code> , <code>admit</code> , or <code>unsafe</code> declaration reported on the active AASC/Hodge audit surface

The verification command supplied by the release is:

```
powershell -ExecutionPolicy Bypass -File scripts/check-hodge-endpoint-audit.ps1
```

The audit runner scans the active AASC/Hodge Lean surface for live project-level `axiom`, `sorry`, `admit`, or `unsafe` declarations, builds the Hodge audit runner module, and evaluates the focused AASC/Hodge axiom-check files. The GitHub workflow `.github/workflows/audit.yml` runs the same audit on push, pull request, and manual dispatch.

F.2 Declaration locations

The manuscript-facing Hodge declarations are concentrated in `MaleyLean/Papers/Hodge/EndpointClosure.lean`. Status and boundary ledgers are recorded in `StatusLedger.lean`, `TruthBoundary.lean`, and `AuditRunners.lean`. The focused axiom checks are under `Checks/Axiom/`.

Lean declaration	Manuscript role
<code>HodgeCycleCarrier</code>	Semantic carrier for a Hodge-standing rational class
<code>HodgeStanding</code>	Hodge-standing predicate for the represented carrier
<code>FixedHodgeCycleCarrier</code>	Fixed Hodge/cycle carrier condition
<code>NonemptyCycleFiber</code>	Positive endpoint-image occupation
<code>EmptyCycleFiber</code>	Bare empty-fiber negative content
<code>OfficialHodgeEndpoint</code>	Represented universal Hodge endpoint
<code>HodgeEndpointUse</code>	Endpoint-use marker
<code>HodgeOfficialEmptyFiberCounterexampleForce</code>	Official endpoint-defeating empty-fiber force
<code>HodgeLocalEmptyFiberCounterexample</code>	Local exact-complement counterexample role
<code>CycleFiberEndpointNonOccupation</code>	Projection of empty-fiber content to cycle-fiber non-occupation
<code>EmptyCycleFiberSeparatorOccupation</code>	Empty-fiber separator endpoint occupation
<code>IndependentFiberDiscriminator</code>	D_{fib} role
<code>NoIndependentFiberDiscriminator</code>	N_{fib} closure

Lean declaration	Manuscript role
NFib, DFib	Lean-side abbreviations for no-independent closure and fiber discriminator
HodgeEndpointStatus	Positive/empty-fiber-separator status surface
EmptyFiberUseKind	Use-kind classifier for bare, theoremhood, bridge, support, book-keeping, domain-shift, discriminator, and autonomous-sixth-role branches

The proof-spine theorem anchors are:

Lean theorem	Manuscript counterpart
<code>hodgeEndpointUse_forces_kernel</code>	endpoint use instantiates the kernel
<code>hodgeEndpointUse_satisfies_targetAdequacy</code>	endpoint use satisfies target adequacy
<code>hodgeEndpointUse_instantiates_aPlus</code>	endpoint use triggers A^+
<code>hodgeBareEmptyFiberContent_preserved</code>	bare content remains lawful content
<code>hodgeOrdinaryEmptyFiberTheoremhood_kernelInternal</code>	ordinary negative theoremhood is kernel-internal
<code>hodgeOfficialEmptyFiberCounterexampleForce_endpointUse</code>	official empty-fiber counterexample force is endpoint use
<code>hodgeNoAutonomousCounterexampleEndpointRole</code>	no autonomous counterexample endpoint role
<code>hodgeNoAutonomousSixthEndpointRole</code>	no sixth empty-fiber endpoint role
<code>hodgeOfficialCounterexampleForce_projects_to_emptyFiberSeparator</code>	official counterexample force projects to empty-fiber separator occupation
<code>hodgeLocalCounterexample_projects_to_emptyFiberSeparator</code>	local counterexample projects to local empty-fiber separator occupation
<code>hodgeEmptyFiberSeparator_induces_DFib</code>	empty-fiber separator induces D_{fib}
<code>hodgeEndpointUse_yields_NFib</code>	endpoint use supplies N_{fib} when no-independent closure is present
<code>hodgeNFib_excludes_DFib</code>	N_{fib} excludes D_{fib}
<code>hodgeLocalCounterexample_impossible_of_noIndependentDiscriminator</code>	local empty-fiber counterexample contradiction
<code>hodgeNonemptyCycleFiber_forced</code>	pointwise nonempty fiber closeout
<code>hodgePointwiseEndpoint_of_aascContext</code>	pointwise endpoint under AASC context
<code>hodgeEndpoint_of_aascContext</code>	represented universal Hodge endpoint
<code>hodgeEndpointClosurePercent_eq</code>	audit-progress equality
	<code>HodgeEndpointClosurePercent = 100</code>

F.3 Manuscript-to-Lean correspondence

The manuscript-to-Lean symbol correspondence is:

Manuscript expression	Lean-facing counterpart
X, p, α fixed Hodge/cycle instance	<code>HodgeCycleCarrier c</code>
$\alpha \in \text{Hdg}^p(X)$	<code>HodgeStanding c</code>
Fixed Hodge/cycle carrier	<code>FixedHodgeCycleCarrier c</code>
$F_X(\alpha) \neq \emptyset$	<code>NonemptyCycleFiber c</code>
$F_X(\alpha) = \emptyset$	<code>EmptyCycleFiber c</code>
Official Hodge endpoint	<code>OfficialHodgeEndpoint</code>
Endpoint use	<code>HodgeEndpointUse c</code>
Bare empty-fiber content	<code>HodgeBareEmptyFiberContent c</code>
Ordinary empty-fiber theoremhood	<code>HodgeOrdinaryEmptyFiberTheoremhood c</code>
Official empty-fiber counterexample force	<code>HodgeOfficialEmptyFiberCounterexampleForce c</code>
Local empty-fiber counterexample	<code>HodgeLocalEmptyFiberCounterexample c</code>
Cycle-fiber non-occupation	<code>CycleFiberEndpointNonOccupation c</code>
Empty-fiber separator occupation	<code>EmptyCycleFiberSeparatorOccupation c</code>
$D_{\text{fib}}(X, p, \alpha)$	<code>DFib c / IndependentFiberDiscriminator c</code>
$N_{\text{fib}}(X, p, \alpha)$	<code>NFib c / NoIndependentFiberDiscriminator c</code>
No autonomous sixth role	<code>EmptyFiberUseKind.autonomousSixthRole</code> has no classification inhabitant
Pointwise closeout	<code>hodgeNonemptyCycleFiber_forced</code>
Universal closeout	<code>hodgeEndpoint_of_aascContext</code>

The formal correspondence of the central proof chain is:

```

HodgeOfficialEmptyFiberCounterexampleForce c
  -> HodgeEndpointUse c
  -> EmptyCycleFiberSeparatorOccupation c
  -> DFib c

```

```

HodgeEndpointUse c
  + NoIndependentFiberDiscriminator c
  -> NFib c
  -> Not (DFib c)

```

```

HodgeEndpointUse c
  + NoIndependentFiberDiscriminator c
  + HodgeStanding c
  + FixedHodgeCycleCarrier c
  -> NonemptyCycleFiber c

```

```

HodgeAASCContext
  -> OfficialHodgeEndpoint

```

F.4 Selected source excerpts

The following excerpts are lightly reflowed from `EndpointClosure.lean` to display the Lean-side route without relying on long repository line structure.

```

structure HodgeCycleCarrier where
  id : Nat
  hodgeStanding : Bool
  fixedCycleCarrier : Bool
  nonemptyCycleFiber : Bool

def HodgeStanding (c : HodgeCycleCarrier) : Prop :=
  c.hodgeStanding = true

def FixedHodgeCycleCarrier (c : HodgeCycleCarrier) : Prop :=
  c.fixedCycleCarrier = true

def NonemptyCycleFiber (c : HodgeCycleCarrier) : Prop :=
  c.nonemptyCycleFiber = true

def EmptyCycleFiber (c : HodgeCycleCarrier) : Prop :=
  Not (NonemptyCycleFiber c)

def OfficialHodgeEndpoint : Prop :=
  forall c : HodgeCycleCarrier,
    HodgeStanding c -> NonemptyCycleFiber c

def HodgeOfficialEmptyFiberCounterexampleForce
  (c : HodgeCycleCarrier) : Prop :=
  HodgeBareEmptyFiberContent c /\ HodgeEndpointUse c

```

```

theorem hodgeOfficialEmptyFiberCounterexampleForce_endpointUse
  {c : HodgeCycleCarrier} :
  HodgeOfficialEmptyFiberCounterexampleForce c ->
  HodgeEndpointUse c := by
  intro h
  exact h.2

theorem hodgeNoAutonomousCounterexampleEndpointRole
  {c : HodgeCycleCarrier} :
  Not (HodgeOfficialEmptyFiberCounterexampleForce c /\
    Not (HodgeEndpointUse c)) := by
  intro h
  exact h.2
  (hodgeOfficialEmptyFiberCounterexampleForce_endpointUse h.1)

theorem hodgeOfficialCounterexampleForce_projects_to_emptyFiberSeparator
  {c : HodgeCycleCarrier} :
  HodgeOfficialEmptyFiberCounterexampleForce c ->
  FixedHodgeCycleCarrier c ->
  EmptyCycleFiberSeparatorOccupation c := by
  intro hForce hFixed
  exact And.intro hForce.2 (And.intro hFixed hForce.1.2)

theorem hodgeEmptyFiberSeparator_induces_DFib
  {c : HodgeCycleCarrier} :
  EmptyCycleFiberSeparatorOccupation c -> DFib c := by
  intro hSep
  exact hodgeStatusWork_induces_independentFiberDiscriminator
    (hodgeEmptyFiberSeparator_induces_statusWork hSep)

theorem hodgeNFib_excludes_DFib {c : HodgeCycleCarrier} :
  NFib c -> Not (DFib c) := by
  intro hNo hD
  exact hNo hD

theorem hodgeLocalCounterexample_impossible_of_noIndependentDiscriminator
  {c : HodgeCycleCarrier} :
  HodgeEndpointUse c ->
  NoIndependentFiberDiscriminator c ->
  Not (HodgeLocalEmptyFiberCounterexample c) := by
  intro hUse hNo hLocal
  exact hNo
  (hodgeEmptyFiberSeparator_induces_DFib
    (hodgeLocalCounterexample_projects_to_emptyFiberSeparator
      (And.intro hLocal hUse)))

theorem hodgeNonemptyCycleFiber_forced {c : HodgeCycleCarrier} :
  HodgeEndpointUse c ->
  NoIndependentFiberDiscriminator c ->
  HodgeStanding c ->

```

```

FixedHodgeCycleCarrier c ->
  NonemptyCycleFiber c := by
intro hUse hNo hStanding hFixed
by_cases hFiber : NonemptyCycleFiber c
· exact hFiber
· exact False.elim
  (hodgeLocalCounterexample_impossible_of_noIndependentDiscriminator
    hUse hNo (And.intro (And.intro hStanding hFiber) hFixed))

structure HodgeAASCContext where
  endpointUse : forall c, HodgeStanding c -> HodgeEndpointUse c
  fixedCarrier : forall c, HodgeStanding c -> FixedHodgeCycleCarrier c
  noIndependentFiberDiscriminator :
    forall c, HodgeStanding c -> NoIndependentFiberDiscriminator c

theorem hodgeEndpoint_of_aascContext
  (ctx : HodgeAASCContext) : OfficialHodgeEndpoint := by
intro c hStanding
exact hodgePointwiseEndpoint_of_aascContext
  (ctx.endpointUse c hStanding)
  (ctx.noIndependentFiberDiscriminator c hStanding)
  hStanding
  (ctx.fixedCarrier c hStanding)

```

F.5 Focused axiom-output surface

The release includes focused axiom-check files for the Hodge endpoint surface and the supporting AASC foundation layer. The release records no live project-level `axiom`, `sorry`, `admit`, or `unsafe` declaration on the active AASC/Hodge audit surface. Standard Lean/classical base dependencies may appear in dependency reports; the audit claim is that no project-specific axiom or incomplete declaration is live in the checked surface.

Focused Lean anchor	Project-level axiom status
<code>hodgeOfficialEmptyFiberCounterexampleForce_endpointUse</code>	no project axiom reported
<code>hodgeNoAutonomousCounterexampleEndpointRole</code>	no project axiom reported
<code>hodgeNoAutonomousSixthEndpointRole</code>	no project axiom reported
<code>hodgeOfficialCounterexampleForce_projects_to_emptyFiberSeparator</code>	no project axiom reported
<code>hodgeEmptyFiberSeparator_induces_DFib</code>	no project axiom reported
<code>hodgeNFib_excludes_DFib</code>	no project axiom reported
<code>hodgeLocalCounterexample_impossible_of_noIndependentDiscriminator</code>	no project axiom reported
<code>hodgeNonemptyCycleFiber_forced</code>	no project axiom reported
<code>hodgeEndpoint_of_aascContext</code>	no project axiom reported
<code>hodgeEndpointClosurePercent_eq</code>	no project axiom reported
<code>hodgeMainAnchorsCount_eq</code>	no project axiom reported
<code>hodgeAuditFileNames_count_eq</code>	no project axiom reported
<code>hodgeBoundaryLedgerCount_eq</code>	no project axiom reported
<code>hodgeAuditRunnerEndpointClosurePercent_eq</code>	no project axiom reported

The focused files named by the status ledger are:

```

Checks/Axiom/MinimalConditionsForAdmissibleConstructionAxiomCheck.lean
Checks/Axiom/NonDegenerateConstructionAndKernelOfAdmissibilityAxiomCheck.lean
Checks/Axiom/HodgeEndpointClosureAxiomCheck.lean
Checks/Axiom/HodgeStatusLedgerAxiomCheck.lean

```

Checks/Axiom/HodgeTruthBoundaryAxiomCheck.lean
 Checks/Axiom/HodgeAuditRunnersAxiomCheck.lean
 Checks/Axiom/HodgeFullStackAASCAXiomCheck.lean

F.6 Audit scope and boundary

The Lean archive formalizes the manuscript-facing AASC endpoint route for the Hodge endpoint: fixed-carrier endpoint use, target-specific empty-fiber separator occupation, same-domain discriminator exclusion, local countercase discharge, and represented endpoint closeout. The final Lean endpoint is `OfficialHodgeEndpoint`; the main closeout theorem is `hodgeEndpoint_of_aascContext`, and the pointwise closeout theorem is `hodgeNonemptyCycleFiber forced`.

The audit boundary is essential. Classical Hodge theory supplies the native carrier and endpoint vocabulary: smooth projective complex varieties, codimension, rational Hodge classes, rational Chow groups, algebraic cycles, and the rational cycle-class map. The Lean archive represents those objects semantically by named carrier and adequacy fields and checks the AASC endpoint-role route over that represented carrier. It does not claim a rebuilt-from-foundations formalization of the surrounding algebraic geometry.

The correct reading is therefore: the manuscript remains the proof object; the Lean archive is a reproducibility audit for the AASC/Hodge endpoint proof spine and its no-autonomous-empty-fiber-role closure.

G Hostile-referee audit expansion and replay matrix

This appendix preserves two full replay families in prose–dependency replay and hostile-reader replay—and records the remaining replay families in matrix form. This is a conservative trim: every load-bearing claim remains visible, the object/use distinction remains explicit, and the same-mode denial burden remains attached to the fixed Hodge/cycle carrier.

G.1 Audit packet 1: Context non-smuggling (dependency replay)

Claim under audit. Hodge context fixes carrier and slots without branch occupation.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.2 Audit packet 2: Model adequacy (dependency replay)

Claim under audit. Readout and counterreadout are identified with nonempty and empty cycle fiber.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.3 Audit packet 3: Counterexample force as endpoint use (dependency replay)

Claim under audit. Official endpoint defeat is endpoint use.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.4 Audit packet 4: No autonomous counterexample role (dependency replay)

Claim under audit. There is no endpoint-defeating branch outside endpoint use.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.5 Audit packet 5: ATS locking (dependency replay)

Claim under audit. The same endpoint act preserves anchor and tensor data.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.6 Audit packet 6: UEAP no-overreporting (dependency replay)

Claim under audit. Support-level content cannot be reported as endpoint readout or endpoint defeat.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.7 Audit packet 7: Single cycle-fiber interface (dependency replay)

Claim under audit. Cycle-fiber occupation is the only positive endpoint-image interface.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.8 Audit packet 8: Native counterforce exhaustion (dependency replay)

Claim under audit. Native empty-fiber force has no fifth role.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.9 Audit packet 9: Separator induction (dependency replay)

Claim under audit. Local separator occupation induces the fiber discriminator.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.10 Audit packet 10: A+ no-independent closure (dependency replay)

Claim under audit. Endpoint use excludes independent same-domain discriminators.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.11 Audit packet 11: Annotation discharge (dependency replay)

Claim under audit. The exact-complement annotation is not an added premise.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.12 Audit packet 12: Universalization (dependency replay)

Claim under audit. Arbitrary instance closure yields the global Hodge endpoint.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.13 Audit packet 13: Surrogate separation (dependency replay)

Claim under audit. Motives and absolute Hodge data are bridge programs unless they supply Chow fiber.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $\text{CH}^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $\text{cl}_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.14 Audit packet 14: Known-case transfer (dependency replay)

Claim under audit. Known bridge completions do not transfer universally without target gating.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.15 Audit packet 15: Local-to-global nonfactorization (dependency replay)

Claim under audit. Local cycle data require a global compatibility certificate.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.16 Audit packet 16: No repair (dependency replay)

Claim under audit. Later cycle discovery creates a new bridge-complete act, not repair of an old one.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.17 Audit packet 17: No selector (dependency replay)

Claim under audit. Canonical representatives cannot decide readout before the fiber is fixed.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.18 Audit packet 18: Incompleteness firewall (dependency replay)

Claim under audit. Independence matters only if it re-enters endpoint standing or trajectory.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.19 Audit packet 19: Ordinary theoremhood protection (dependency replay)

Claim under audit. Negative theoremhood is proof legitimacy, not forbidden endpoint governance.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.20 Audit packet 20: Non-explosiveness (dependency replay)

Claim under audit. Empty fiber is lawful until endpoint force is added.

Dependency path. This packet belongs to the dependency replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.21 Audit packet 21: Context non-smuggling (hostile-reader replay)

Claim under audit. Hodge context fixes carrier and slots without branch occupation.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.22 Audit packet 22: Model adequacy (hostile-reader replay)

Claim under audit. Readout and counterreadout are identified with nonempty and empty cycle fiber.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.23 Audit packet 23: Counterexample force as endpoint use (hostile-reader replay)

Claim under audit. Official endpoint defeat is endpoint use.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.24 Audit packet 24: No autonomous counterexample role (hostile-reader replay)

Claim under audit. There is no endpoint-defeating branch outside endpoint use.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.25 Audit packet 25: ATS locking (hostile-reader replay)

Claim under audit. The same endpoint act preserves anchor and tensor data.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.26 Audit packet 26: UEAP no-overreporting (hostile-reader replay)

Claim under audit. Support-level content cannot be reported as endpoint readout or endpoint defeat.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.27 Audit packet 27: Single cycle-fiber interface (hostile-reader replay)

Claim under audit. Cycle-fiber occupation is the only positive endpoint-image interface.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.28 Audit packet 28: Native counterforce exhaustion (hostile-reader replay)

Claim under audit. Native empty-fiber force has no fifth role.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.29 Audit packet 29: Separator induction (hostile-reader replay)

Claim under audit. Local separator occupation induces the fiber discriminator.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.30 Audit packet 30: A+ no-independent closure (hostile-reader replay)

Claim under audit. Endpoint use excludes independent same-domain discriminators.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.31 Audit packet 31: Annotation discharge (hostile-reader replay)

Claim under audit. The exact-complement annotation is not an added premise.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.32 Audit packet 32: Universalization (hostile-reader replay)

Claim under audit. Arbitrary instance closure yields the global Hodge endpoint.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.33 Audit packet 33: Surrogate separation (hostile-reader replay)

Claim under audit. Motives and absolute Hodge data are bridge programs unless they supply Chow fiber.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.34 Audit packet 34: Known-case transfer (hostile-reader replay)

Claim under audit. Known bridge completions do not transfer universally without target gating.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.35 Audit packet 35: Local-to-global nonfactorization (hostile-reader replay)

Claim under audit. Local cycle data require a global compatibility certificate.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.36 Audit packet 36: No repair (hostile-reader replay)

Claim under audit. Later cycle discovery creates a new bridge-complete act, not repair of an old one.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.37 Audit packet 37: No selector (hostile-reader replay)

Claim under audit. Canonical representatives cannot decide readout before the fiber is fixed.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.38 Audit packet 38: Incompleteness firewall (hostile-reader replay)

Claim under audit. Independence matters only if it re-enters endpoint standing or trajectory.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.39 Audit packet 39: Ordinary theoremhood protection (hostile-reader replay)

Claim under audit. Negative theoremhood is proof legitimacy, not forbidden endpoint governance.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.40 Audit packet 40: Non-explosiveness (hostile-reader replay)

Claim under audit. Empty fiber is lawful until endpoint force is added.

Dependency path. This packet belongs to the hostile-reader replay. It depends on the official Hodge route, the fixed cycle-fiber endpoint image, the object/use distinction, and the kernel/A+ endpoint-use machinery. It is never used before the carrier and branch slots are fixed.

Hostile-referee risk. A hostile reader may attempt to treat the relevant content as raw mathematical syntax, ordinary negative theoremhood, surrogate support, or a classical counterexample object rather than endpoint-use material.

Closure and denial burden. The closure response is role-sensitive. If the content is raw, support, bookkeeping, or outside the fixed carrier, it has no endpoint-defeating force. If it keeps endpoint-defeating force on the same carrier, it enters endpoint-use discipline and route exhaustion. A successful objection must preserve the same X, p, α , the same Hodge-standing predicate, the same Chow carrier $CH^p(X)_{\mathbb{Q}}$, and the same rational cycle-class map $cl_{\mathbb{Q}}$.

Use in the proof spine. The packet is reused only after the local exact-complement assumption has been opened. It marks a role transition, not a hidden assertion of nonempty fiber.

G.41 Replay matrix for remaining audit modes

The following matrix extends the forty prose packets above across the remaining replay modes without repeating the same closure paragraphs. For every claim–lens pair, the same fixed data must be preserved: X, p, α , the rational Hodge-standing predicate, $\text{CH}^p(X)_{\mathbb{Q}}$, $\text{cl}_{\mathbb{Q}}$, and $F_X(\alpha)$. A route outside that carrier is a domain shift; a route with no endpoint effect is support or bookkeeping; a same-carrier route with endpoint-defeating force enters endpoint-use discipline and native empty-fiber exhaustion.

Replay lens	What it checks for all twenty audit claims
same-mode denial replay endpoint-use replay	the objection must preserve the fixed carrier while denying a named theorem link. endpoint-defeating force is tested as endpoint use, while raw/support content remains below endpoint use.
Lean-audit replay	the claim is checked against the Lean endpoint-role audit surface when Lean support is invoked.
route-exhaustion replay	the route must be bridge-neutralized, support-only, bookkeeping, domain shift, or discriminator-inducing.
anti-circularity replay counterexample-force replay	the claim must not assert nonempty fiber before the local closeout. a proposed counterexample object is separated from the act of deploying it as official endpoint defeat.

Audit claim	Standing denial burden retained in the matrix
Context non-smuggling	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Model adequacy	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Counterexample force as endpoint use	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
No autonomous counterexample role	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
ATS locking	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
UEAP no-overreporting	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Single cycle-fiber interface	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Native counterforce exhaustion	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Separator induction	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
A+ no-independent closure	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Annotation discharge	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Universalization	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Surrogate separation	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Known-case transfer	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .

Audit claim	Standing denial burden retained in the matrix
Local-to-global nonfactorization	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
No repair	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
No selector	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Incompleteness firewall	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Ordinary theoremhood protection	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .
Non-explosiveness	The critic must preserve the fixed Hodge/cycle carrier and break this claim under at least one replay lens without reducing to support, bookkeeping, domain shift, bridge completion, or D_{fib} governed by N_{fib} .

H Faithful counterexample worksheet

This worksheet states the concrete burden for a successful same-mode counterexample. Each proposed objection must remain on the fixed Hodge/cycle carrier. If it changes the carrier, it is a domain shift. If it changes no endpoint feature, it is support or bookkeeping. If it preserves endpoint-defeating force, it falls under endpoint-use and native-counterforce exhaustion.

Objection target	Successful objection must produce	Failure mode if it does not
Autonomous endpoint counterexample	endpoint-defeating empty-fiber force that is not endpoint use	raw/support/bookkeeping if it lacks force; domain shift if carrier changes; otherwise endpoint use.
Fifth native role	same-carrier empty-fiber endpoint force outside bridge-neutralized, support-only, bookkeeping, domain shift, and D_{fib}	native counterforce exhaustion collapses the role.
Surrogate occupation	a non-Chow surrogate that occupies $F_X(\alpha)$ without bridge	support or domain shift unless it supplies the Chow fiber.
Selector authority	canonical representative selection that decides readout before fiber occupation	AMetric no-selector failure or bookkeeping.
Repair	later bridge data repairing an earlier unbridged report as the same act	irreversibility; the later bridge-complete report is a new act.
Local factorization	local cycle data becoming a global Chow fiber without compatibility certificate	support until compatibility and global-fiber certificate.
A^+ failure	same-domain independent discriminator compatible with no-independent closure	contradicts $N_{\text{fib}} \Rightarrow \neg D_{\text{fib}}$.
Projection failure	algebraic readout not equivalent to nonempty cycle-fiber status	changes the official Hodge/cycle endpoint carrier.
Annotation failure	local annotation as an added object-level premise	annotation is proof-role metadata; only $[F_X(\alpha) = \emptyset]_i$ is discharged.
Universalization failure	fixed-instance nonempty fiber failing to yield the universal endpoint	must block ordinary universal generalization over arbitrary X, p, α .

I Expanded stress-test capsules and report-preservation matrix

This appendix preserves the endpoint-force and route-exhaustion stress-test capsules in full prose. The report-preservation lens is then recorded in matrix form. The raw Hodge-theoretic content remains lawful

throughout; the audit concerns only endpoint-use role, fixed-carrier preservation, report level, and the route from empty-fiber endpoint counterforce to the no-independent-discriminator contradiction.

I.1 Stress test: Official endpoint identity (endpoint-force lens)

Load-bearing claim. This stress test audits the identification of the Clay-facing endpoint with universal nonempty rational cycle-class fibers. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.2 Stress test: Context non-smuggling (endpoint-force lens)

Load-bearing claim. This stress test audits the claim that the Hodge endpoint context fixes the carrier and slots without asserting nonempty fiber. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.3 Stress test: Hodge-standing exactness (endpoint-force lens)

Load-bearing claim. This stress test audits the representation of rational Hodge standing as cohomological type status. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.4 Stress test: Cycle-fiber exactness (endpoint-force lens)

Load-bearing claim. This stress test audits the representation of algebraic readout by the fiber $F_X(\alpha)$. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.5 Stress test: Empty-fiber exactness (endpoint-force lens)

Load-bearing claim. This stress test audits the negative local branch as fiber non-occupation. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.6 Stress test: Counterexample object/use distinction (endpoint-force lens)

Load-bearing claim. This stress test audits the separation between an object satisfying the negative branch and the act of offering it as endpoint defeat. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.7 Stress test: Official counterexample force (endpoint-force lens)

Load-bearing claim. This stress test audits the nonclassical dependency identifying endpoint-defeating counterexample force with theorem-bearing endpoint use. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit.

It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.8 Stress test: No autonomous counterexample endpoint role (endpoint-force lens)

Load-bearing claim. This stress test audits the denial of a fifth role outside endpoint use for official endpoint defeat. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.9 Stress test: Kernel forcing (endpoint-force lens)

Load-bearing claim. This stress test audits the derivation of reference, standing, admissibility, and irreversibility from target adequacy. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape

that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.10 Stress test: A+ consequence layer (endpoint-force lens)

Load-bearing claim. This stress test audits the fixed-domain consequence package binding endpoint use. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.11 Stress test: ATS locking (endpoint-force lens)

Load-bearing claim. This stress test audits same target, carrier, branch, readout trajectory, and no second classifier. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.12 Stress test: UEAP report discipline (endpoint-force lens)

Load-bearing claim. This stress test audits no report exceeding carrier, bridge, and support status. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.13 Stress test: AMetric no-selector boundary (endpoint-force lens)

Load-bearing claim. This stress test audits no preferred representative or scalar ranking as pre-bridge authority. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.14 Stress test: Surrogate carrier separation (endpoint-force lens)

Load-bearing claim. This stress test audits motivic, variational, or obstruction objects as bridge support rather than readout. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.15 Stress test: No post-selection repair (endpoint-force lens)

Load-bearing claim. This stress test audits later cycle discovery as fresh bridge completion rather than same-act repair. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.16 Stress test: No carrier transfer (endpoint-force lens)

Load-bearing claim. This stress test audits known cases and neighboring domains not carrying endpoint standing without target inclusion. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.17 Stress test: No generators (endpoint-force lens)

Load-bearing claim. This stress test audits Hodge standing not generating Chow-fiber nonemptiness. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.18 Stress test: No silent redescription (endpoint-force lens)

Load-bearing claim. This stress test audits no replacement of Chow-fiber readout by another carrier under same label. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.19 Stress test: Non-factorizability (endpoint-force lens)

Load-bearing claim. This stress test audits local bridge fragments not composing into global readout without compatibility. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.20 Stress test: Role-occupancy closure (endpoint-force lens)

Load-bearing claim. This stress test audits nonempty fiber as role occupation rather than unique representative selection. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.21 Stress test: Cycle-fiber single interface (endpoint-force lens)

Load-bearing claim. This stress test audits the single endpoint-image interface for algebraic readout. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.22 Stress test: Coequal carrier distinction (endpoint-force lens)

Load-bearing claim. This stress test audits separating broad outcome carrier from cycle-fiber image carrier. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.23 Stress test: Native empty-fiber exhaustion (endpoint-force lens)

Load-bearing claim. This stress test audits bridge-neutralized, support-only, bookkeeping, domain shift, or discriminator. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.24 Stress test: Bridge-neutralized branch (endpoint-force lens)

Load-bearing claim. This stress test audits why a bridge-neutralized negative branch is not live endpoint counterforce. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.25 Stress test: Support-only branch (endpoint-force lens)

Load-bearing claim. This stress test audits why support status cannot defeat endpoint closure. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.26 Stress test: Bookkeeping branch (endpoint-force lens)

Load-bearing claim. This stress test audits why labels and ledgers do not determine endpoint status. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.27 Stress test: Domain-shift branch (endpoint-force lens)

Load-bearing claim. This stress test audits why changing X, p, α , coefficients, equivalence, or carrier exits the fixed endpoint. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.28 Stress test: Cycle-fiber discriminator (endpoint-force lens)

Load-bearing claim. This stress test audits the independent same-domain discriminator induced by endpoint-standing empty-fiber force. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.29 Stress test: No-independent closure (endpoint-force lens)

Load-bearing claim. This stress test audits the A+ consequence excluding the independent discriminator. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.30 Stress test: Local separator occupation (endpoint-force lens)

Load-bearing claim. This stress test audits the local empty-fiber counterclaim as endpoint-defeating non-occupation. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.31 Stress test: Reductio annotation (endpoint-force lens)

Load-bearing claim. This stress test audits proof-act annotation not object-level premise. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.32 Stress test: Annotation discharge (endpoint-force lens)

Load-bearing claim. This stress test audits the original empty-fiber assumption is discharged. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.33 Stress test: Ordinary theoremhood protection (endpoint-force lens)

Load-bearing claim. This stress test audits negative theoremhood is allowed until used as official endpoint resolution. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.34 Stress test: Incompleteness firewall (endpoint-force lens)

Load-bearing claim. This stress test audits same-domain independence matters only if endpoint-standing relevant. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.35 Stress test: Route-locus exhaustion (endpoint-force lens)

Load-bearing claim. This stress test audits status, codomain, existence, carrier, temporal, locality, selector, and second-classifier loci. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.36 Stress test: Weakening resistance (endpoint-force lens)

Load-bearing claim. This stress test audits weaker regimes preserve behavior, shift domain, lower status, or reintroduce forbidden governance. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.37 Stress test: Official endpoint identity (route-exhaustion lens)

Load-bearing claim. This stress test audits the identification of the Clay-facing endpoint with universal nonempty rational cycle-class fibers. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.38 Stress test: Context non-smuggling (route-exhaustion lens)

Load-bearing claim. This stress test audits the claim that the Hodge endpoint context fixes the carrier and slots without asserting nonempty fiber. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.39 Stress test: Hodge-standing exactness (route-exhaustion lens)

Load-bearing claim. This stress test audits the representation of rational Hodge standing as cohomological type status. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.40 Stress test: Cycle-fiber exactness (route-exhaustion lens)

Load-bearing claim. This stress test audits the representation of algebraic readout by the fiber $F_X(\alpha)$. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.41 Stress test: Empty-fiber exactness (route-exhaustion lens)

Load-bearing claim. This stress test audits the negative local branch as fiber non-occupation. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.42 Stress test: Counterexample object/use distinction (route-exhaustion lens)

Load-bearing claim. This stress test audits the separation between an object satisfying the negative branch and the act of offering it as endpoint defeat. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.43 Stress test: Official counterexample force (route-exhaustion lens)

Load-bearing claim. This stress test audits the nonclassical dependency identifying endpoint-defeating counterexample force with theorem-bearing endpoint use. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit.

It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.44 Stress test: No autonomous counterexample endpoint role (route-exhaustion lens)

Load-bearing claim. This stress test audits the denial of a fifth role outside endpoint use for official endpoint defeat. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.45 Stress test: Kernel forcing (route-exhaustion lens)

Load-bearing claim. This stress test audits the derivation of reference, standing, admissibility, and irreversibility from target adequacy. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape

that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.46 Stress test: A+ consequence layer (route-exhaustion lens)

Load-bearing claim. This stress test audits the fixed-domain consequence package binding endpoint use. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.47 Stress test: ATS locking (route-exhaustion lens)

Load-bearing claim. This stress test audits same target, carrier, branch, readout trajectory, and no second classifier. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.48 Stress test: UEAP report discipline (route-exhaustion lens)

Load-bearing claim. This stress test audits no report exceeding carrier, bridge, and support status. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.49 Stress test: AMetric no-selector boundary (route-exhaustion lens)

Load-bearing claim. This stress test audits no preferred representative or scalar ranking as pre-bridge authority. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.50 Stress test: Surrogate carrier separation (route-exhaustion lens)

Load-bearing claim. This stress test audits motivic, variational, or obstruction objects as bridge support rather than readout. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.51 Stress test: No post-selection repair (route-exhaustion lens)

Load-bearing claim. This stress test audits later cycle discovery as fresh bridge completion rather than same-act repair. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.52 Stress test: No carrier transfer (route-exhaustion lens)

Load-bearing claim. This stress test audits known cases and neighboring domains not carrying endpoint standing without target inclusion. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.53 Stress test: No generators (route-exhaustion lens)

Load-bearing claim. This stress test audits Hodge standing not generating Chow-fiber nonemptiness. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.54 Stress test: No silent redescription (route-exhaustion lens)

Load-bearing claim. This stress test audits no replacement of Chow-fiber readout by another carrier under same label. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.55 Stress test: Non-factorizability (route-exhaustion lens)

Load-bearing claim. This stress test audits local bridge fragments not composing into global readout without compatibility. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.56 Stress test: Role-occupancy closure (route-exhaustion lens)

Load-bearing claim. This stress test audits nonempty fiber as role occupation rather than unique representative selection. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.57 Stress test: Cycle-fiber single interface (route-exhaustion lens)

Load-bearing claim. This stress test audits the single endpoint-image interface for algebraic readout. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.58 Stress test: Coequal carrier distinction (route-exhaustion lens)

Load-bearing claim. This stress test audits separating broad outcome carrier from cycle-fiber image carrier. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.59 Stress test: Native empty-fiber exhaustion (route-exhaustion lens)

Load-bearing claim. This stress test audits bridge-neutralized, support-only, bookkeeping, domain shift, or discriminator. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.60 Stress test: Bridge-neutralized branch (route-exhaustion lens)

Load-bearing claim. This stress test audits why a bridge-neutralized negative branch is not live endpoint counterforce. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.61 Stress test: Support-only branch (route-exhaustion lens)

Load-bearing claim. This stress test audits why support status cannot defeat endpoint closure. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.62 Stress test: Bookkeeping branch (route-exhaustion lens)

Load-bearing claim. This stress test audits why labels and ledgers do not determine endpoint status. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.63 Stress test: Domain-shift branch (route-exhaustion lens)

Load-bearing claim. This stress test audits why changing X, p, α , coefficients, equivalence, or carrier exits the fixed endpoint. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.64 Stress test: Cycle-fiber discriminator (route-exhaustion lens)

Load-bearing claim. This stress test audits the independent same-domain discriminator induced by endpoint-standing empty-fiber force. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.65 Stress test: No-independent closure (route-exhaustion lens)

Load-bearing claim. This stress test audits the A+ consequence excluding the independent discriminator. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.66 Stress test: Local separator occupation (route-exhaustion lens)

Load-bearing claim. This stress test audits the local empty-fiber counterclaim as endpoint-defeating non-occupation. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.67 Stress test: Reductio annotation (route-exhaustion lens)

Load-bearing claim. This stress test audits proof-act annotation not object-level premise. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.68 Stress test: Annotation discharge (route-exhaustion lens)

Load-bearing claim. This stress test audits the original empty-fiber assumption is discharged. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.69 Stress test: Ordinary theoremhood protection (route-exhaustion lens)

Load-bearing claim. This stress test audits negative theoremhood is allowed until used as official endpoint resolution. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.70 Stress test: Incompleteness firewall (route-exhaustion lens)

Load-bearing claim. This stress test audits same-domain independence matters only if endpoint-standing relevant. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.71 Stress test: Route-locus exhaustion (route-exhaustion lens)

Load-bearing claim. This stress test audits status, codomain, existence, carrier, temporal, locality, selector, and second-classifier loci. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.72 Stress test: Weakening resistance (route-exhaustion lens)

Load-bearing claim. This stress test audits weaker regimes preserve behavior, shift domain, lower status, or reintroduce forbidden governance. The claim is not allowed to function by implication, atmosphere, or familiarity with AASC vocabulary. It must be available as a same-carrier theorem target inside the Hodge endpoint proof and must be reused only after the carrier, branch, and report slots have been instantiated.

Hostile reconstruction. A hostile referee will try to reconstruct this point in the strongest ordinary Hodge-theoretic language: rational Hodge classes are native cohomological objects, Chow groups and cycle-class maps are ordinary algebraic-geometric carriers, and an empty fiber would be ordinary mathematical falsification. The audit therefore never answers by declaring the underlying mathematical content illicit. It answers only by separating raw content, support status, theoremhood, endpoint force, and the fixed cycle-fiber endpoint image.

Closure condition. The claim survives this stress test only if it preserves the same X , p , and α , the same rational Hodge-standing predicate, the same group $\mathrm{CH}^p(X)_{\mathbb{Q}}$, the same map $\mathrm{cl}_{\mathbb{Q}}$, and the same fiber $F_X(\alpha)$. A proposed escape that changes any of these is a domain or carrier shift. A proposed escape that preserves them while retaining endpoint-defeating force enters endpoint-use discipline and must pass through the native empty-fiber exhaustion.

Audit result. Under the fixed endpoint assumptions, the hostile reconstruction leaves only the declared branches: bridge completion, support-only content, bookkeeping, explicit domain shift, or independent same-domain cycle-fiber endpoint discrimination. This packet is therefore not decorative; it is a repeatable closure capsule for the theorem ladder and denial-burden table.

I.73 Report-preservation lens matrix

The report-preservation lens is common across the stress-test topics. It asks whether the report stays within its carrier, bridge support, theoremhood status, and endpoint-use role. A report that exceeds those constraints is overreporting; a report that changes the carrier is domain shift; a report that preserves same-carrier endpoint-defeating force enters endpoint-use discipline.

Stress-test topic	Report-preservation reading
Official endpoint identity	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Context non-smuggling	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Hodge-standing exactness	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Cycle-fiber exactness	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Empty-fiber exactness	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Counterexample object/use distinction	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Official counterexample force	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
No autonomous counterexample endpoint role	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Kernel forcing	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
A+ consequence layer	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
ATS locking	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
UEAP report discipline	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
AMetric no-selector boundary	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Surrogate carrier separation	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
No post-selection repair	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
No carrier transfer	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
No generators	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
No silent redescription	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Non-factorizability	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Role-occupancy closure	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Cycle-fiber single interface	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Coequal carrier distinction	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Native empty-fiber exhaustion	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Bridge-neutralized branch	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Support-only branch	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Bookkeeping branch	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.

Stress-test topic	Report-preservation reading
Domain-shift branch	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Cycle-fiber discriminator	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
No-independent closure	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Local separator occupation	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Reductio annotation	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Annotation discharge	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Ordinary theoremhood protection	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Incompleteness firewall	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Route-locus exhaustion	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.
Weakening resistance	The report must not exceed the fixed Hodge/cycle carrier, the available bridge support, and the actual proof-role status of the content.

References

- [1] Clay Mathematics Institute, *Hodge Conjecture*, official Millennium Problem statement.
- [2] Claire Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge University Press, 2002.
- [3] Claire Voisin, *Hodge Theory and Complex Algebraic Geometry II*, Cambridge University Press, 2003.
- [4] Phillip Griffiths and Joseph Harris, *Principles of Algebraic Geometry*, Wiley, 1978.
- [5] William Fulton, *Intersection Theory*, second edition, Springer, 1998.
- [6] James D. Lewis, *A Survey of the Hodge Conjecture*, American Mathematical Society, 1999.
- [7] Eduardo Cattani, Pierre Deligne, and Aroldo Kaplan, *On the locus of Hodge classes*, Journal of the American Mathematical Society 8 (1995), 483–506.
- [8] Yves Andre, *Une introduction aux motifs*, Societe Mathematique de France, 2004.
- [9] Pierre Deligne, *Hodge cycles on abelian varieties*, Lecture Notes in Mathematics 900, Springer, 1982.
- [10] Steven L. Kleiman, *The standard conjectures*, Proceedings of Symposia in Pure Mathematics 55, American Mathematical Society, 1994.
- [11] Amos Jay Maley, *Non-Degenerate Construction and the Kernel of Admissibility*, 2026.
- [12] Amos Jay Maley, *Claim Standing and Legitimacy under UEAP*, 2026.
- [13] Amos Jay Maley, *Anchor, Tensor, and Skin*, 2026.
- [14] Amos Jay Maley, *The AMetric Boundary*, 2026.
- [15] Amos Jay Maley, *Role-Occupancy Closure under Admissibility*, 2026.
- [16] Amos Jay Maley, *Admissible Construction and the Non-Instantiability of Godel Architectures*, 2026.
- [17] Amos Jay Maley, *AASC-Hodge-Endpoint-Lean-Audit*, GitHub repository, 2026. <https://github.com/somamaley-ux/AASC-Hodge-Endpoint-Lean-Audit>.

-
- [18] Amos Jay Maley, *AASC Hodge Endpoint Lean Audit Archive*, Zenodo, 2026. DOI: <https://doi.org/10.5281/zenodo.20640238>.